

Name: \_\_\_\_\_

## GCSE Physics Unit 1 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

| Year | Score /80 | Grade | Good Questions | Bad Questions | Reasons |
|------|-----------|-------|----------------|---------------|---------|
| 2017 |           |       |                |               |         |
| 2018 |           |       |                |               |         |
| 2019 |           |       |                |               |         |
| 2022 |           |       |                |               |         |
| 2023 |           |       |                |               |         |

Grade boundaries - Raw Score

| Year | E  | D  | C  | B  | A  | A* | Max |
|------|----|----|----|----|----|----|-----|
| 2017 | 6  | 10 | 17 | 24 | 32 | 40 | 48  |
| 2018 | 11 | 13 | 16 | 24 | 32 | 40 | 49  |
| 2019 | 8  | 12 | 20 | 28 | 36 | 44 | 52  |
| 2022 | 4  | 6  | 13 | 22 | 31 | 40 | 49  |
| 2023 | 5  | 10 | 17 | 25 | 34 | 48 | 52  |



|             |               |                  |
|-------------|---------------|------------------|
| Surname     | Centre Number | Candidate Number |
| Other Names |               | 0                |



**GCSE – NEW**

3420UA0-1



**PHYSICS – Unit 1:  
Electricity, Energy and Waves**

**HIGHER TIER**

**MONDAY, 19 JUNE 2017 – MORNING**

**1 hour 45 minutes**

| For Examiner's use only |              |              |
|-------------------------|--------------|--------------|
| Question                | Maximum Mark | Mark Awarded |
| 1.                      | 8            |              |
| 2.                      | 12           |              |
| 3.                      | 10           |              |
| 4.                      | 9            |              |
| 5.                      | 16           |              |
| 6.                      | 13           |              |
| 7.                      | 12           |              |
| Total                   | 80           |              |

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**ADDITIONAL MATERIALS**

In addition to this paper you may require a calculator, a ruler and a drawing compass.

**INSTRUCTIONS TO CANDIDATES**

- Use black ink or black ball-point pen.
- Write your name, centre number and candidate number in the spaces at the top of this page.
- Answer **all** questions.
- Write your answers in the spaces provided in this booklet.

**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.  
Question **6(a)** is a quality of extended response (QER) question where your writing skills will be assessed.

## Equations

|                                                                                                                                                                                      |                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| current = $\frac{\text{voltage}}{\text{resistance}}$                                                                                                                                 | $I = \frac{V}{R}$                              |
| total resistance in a series circuit                                                                                                                                                 | $R = R_1 + R_2$                                |
| total resistance in a parallel circuit                                                                                                                                               | $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$  |
| energy transferred = power $\times$ time                                                                                                                                             | $E = Pt$                                       |
| power = voltage $\times$ current                                                                                                                                                     | $P = VI$                                       |
| power = current <sup>2</sup> $\times$ resistance                                                                                                                                     | $P = I^2R$                                     |
| % efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$                                                            |                                                |
| density = $\frac{\text{mass}}{\text{volume}}$                                                                                                                                        | $\rho = \frac{m}{V}$                           |
| units used (kWh) = power (kW) $\times$ time (h)<br>cost = units used $\times$ cost per unit                                                                                          |                                                |
| wave speed = wavelength $\times$ frequency                                                                                                                                           | $v = \lambda f$                                |
| speed = $\frac{\text{distance}}{\text{time}}$                                                                                                                                        |                                                |
| pressure = $\frac{\text{force}}{\text{area}}$                                                                                                                                        | $p = \frac{F}{A}$                              |
| $p$ = pressure<br>$V$ = volume<br>$T$ = kelvin temperature                                                                                                                           | $\frac{pV}{T} = \text{constant}$               |
|                                                                                                                                                                                      | $T / \text{K} = \theta / ^\circ\text{C} + 273$ |
| change in thermal energy = mass $\times$ specific heat capacity $\times$ change in temperature                                                                                       | $\Delta Q = mc\Delta\theta$                    |
| thermal energy for a change of state = mass $\times$ specific latent heat                                                                                                            | $Q = mL$                                       |
| force on a conductor (at right angles to a magnetic field) carrying a current = magnetic field strength $\times$ current $\times$ length                                             | $F = BIl$                                      |
| $V_1$ = voltage across the primary coil<br>$V_2$ = voltage across the secondary coil<br>$N_1$ = number of turns on the primary coil<br>$N_2$ = number of turns on the secondary coil | $\frac{V_1}{V_2} = \frac{N_1}{N_2}$            |

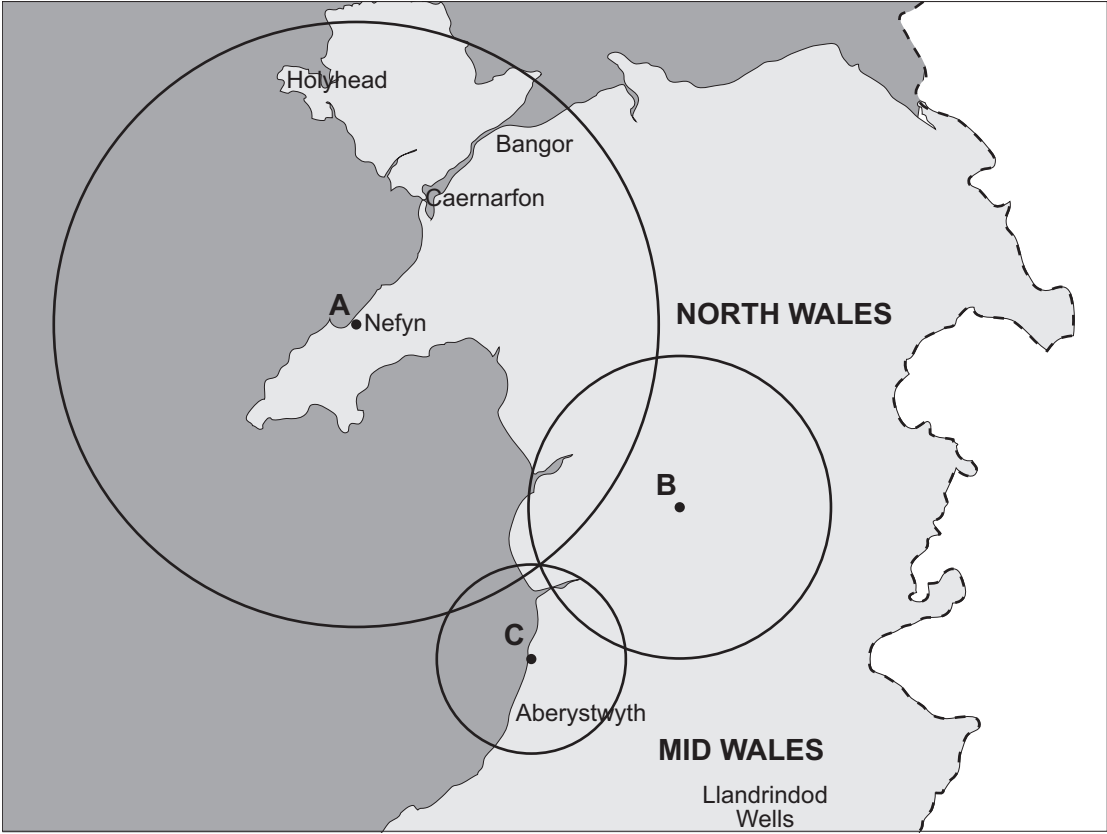
## SI multipliers

| Prefix | Multiplier |
|--------|------------|
| p      | $10^{-12}$ |
| n      | $10^{-9}$  |
| $\mu$  | $10^{-6}$  |
| m      | $10^{-3}$  |

| Prefix | Multiplier |
|--------|------------|
| k      | $10^3$     |
| M      | $10^6$     |
| G      | $10^9$     |
| T      | $10^{12}$  |

Answer all questions.

1. The waves from an earthquake can be monitored by seismological stations located on the surface of the Earth.



The diagram shows circles drawn around three seismological stations **A**, **B** and **C** in Mid and North Wales. The radius of each circle shows the distance of the epicentre from each station.

- (a) (i) **Label on the diagram** with a cross (X) the location of the epicentre. [1]
- (ii) Explain why at least three seismological stations are needed to locate the epicentre. [2]

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- (b) Explain which station (**A**, **B** or **C**) detected the earthquake first. [2]

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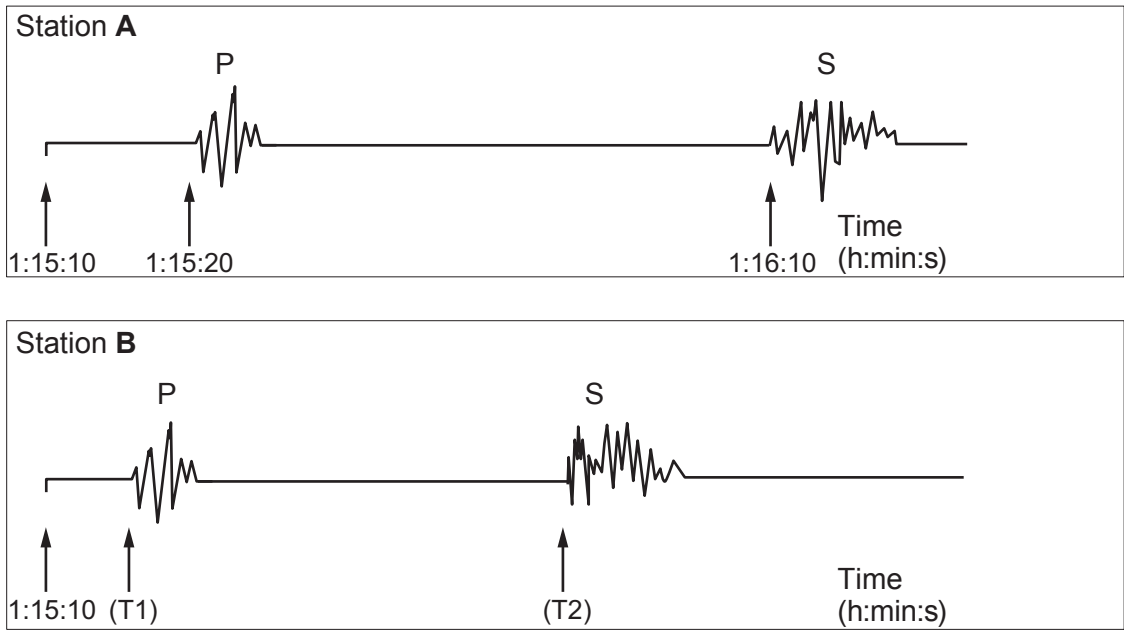
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- (c) The diameter of the circle around station A **is double** the diameter of the circle around station B.

The earthquake happened at 1 hour : 15 minutes : 10 seconds (1:15:10) in the morning.

The following signals were obtained at stations **A** and **B**.



Due to an error on the equipment the **arrival times** (T1 and T2) for the two signals were not recorded at station **B**.

Using signal information from station **A**, determine the times that the P waves and the S waves arrived at station **B**. Explain how you arrived at your answers. [3]

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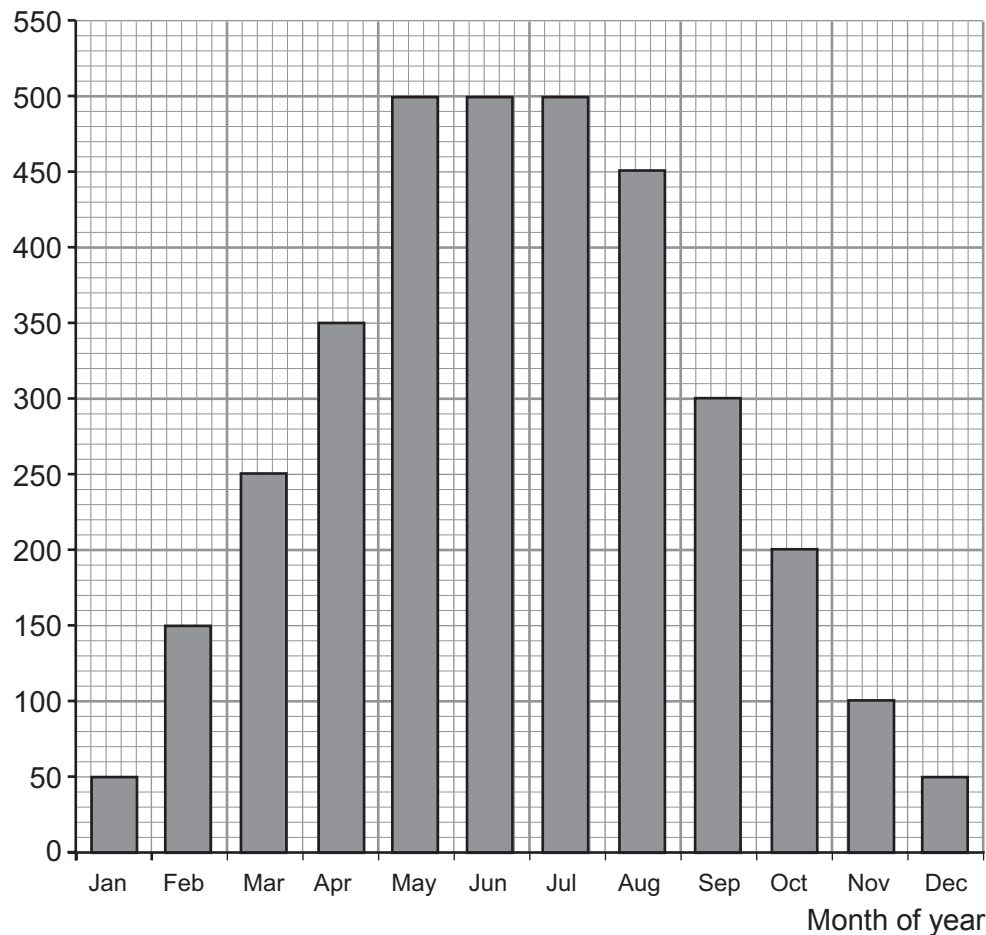
Arrival time of P waves (T1) = ..... hours ..... minutes ..... seconds.

Arrival time of S waves (T2) = ..... hours ..... minutes ..... seconds.

2. A family from Wales researched photovoltaic (PV) cells before they bought some. They were told by the manufacturer that they could expect the PV cells to generate 3400 kWh of electricity per year. **Figure 1** shows the energy generated in kWh from PV cells in a typical year in Wales. This was included in an advert for the PV cells.

**Figure 1**

Energy generated (kWh)



- (a) Use the information from **Figure 1** to validate the claim that PV cells could generate 3400 kWh annually. [2]

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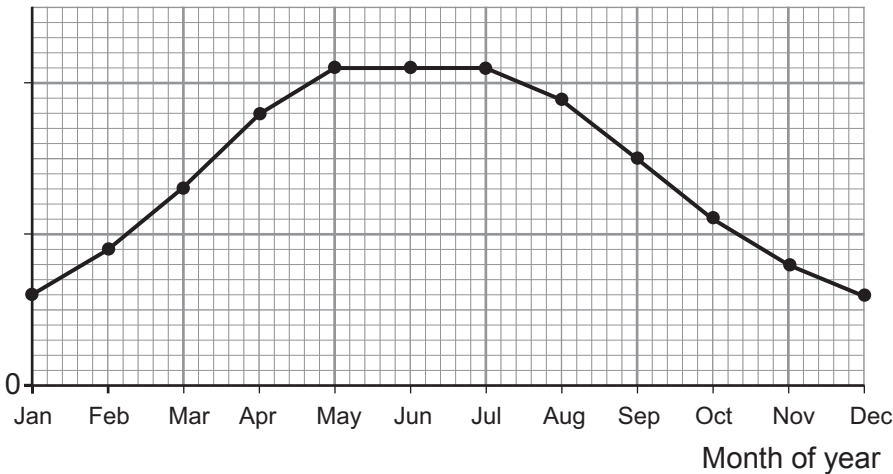
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- (b) Sunshine data for Wales over the last 30 years was obtained from the Welsh MET weather office. **Figure 2** shows the mean number of sunshine hours for each month in Wales.

**Figure 2**

Mean number of  
sunshine hours



- (i) Describe how the information from **Figure 2** supports the general trend shown in **Figure 1**. [1]

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- (ii) The Welsh MET weather office also had data for the **maximum** number of sunshine hours for each month. State why the manufacturers were **not** allowed to use this data in their advert. [1]

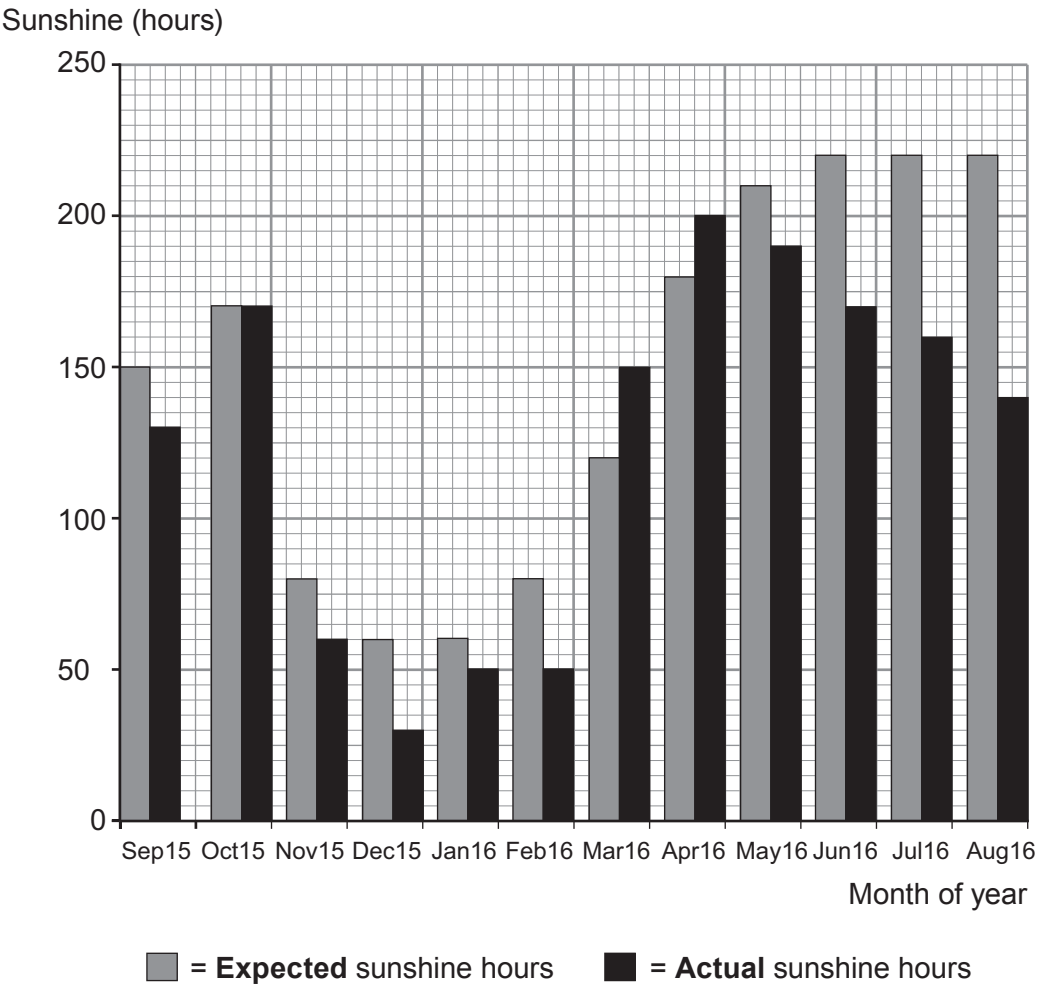
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- (c) In September 2015 the family had PV cells installed on their south facing roof. They collected 12 months of actual data from their PV cells. **Figure 3** compares the **expected** sunshine hours with **actual** sunshine hours. The **expected** sunshine hours data were obtained from local weather station records.

**Figure 3**



- (i) Use information from **Figure 3** to tick (✓) the **three** correct statements. [3]

In Dec15 the **actual** number of sunshine hours was double the **expected** number.

☐

Apr 16 was the month that had the most **actual** sunshine hours.

☐

June, July and August have different **expected** sunshine hours.

☐

Jan 16 had the least number of **actual** sunshine hours.

☐

Apr 16 had 4 times the **actual** number of sunshine hours compared to Jan 16.

☐

There is only one month where the **actual** and the **expected** sunshine hours were the same.

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- (ii) In April 16 the PV cells generated 600 kWh. The manufacturers claim the PV cells will produce 3 kW in sunshine. Use an equation from page 2 and **Figure 3** to validate this claim. [2]

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- (d) The PV cells were expected to produce 3 400 kWh of electricity. However, the total number produced was 3 670 kWh.

- (i) The family saves 29 p for each kWh of energy generated. Calculate how much extra money they saved. [2]

Extra savings = ..... p

- (ii) If the PV cells continue to generate more energy than expected what effect would this have on their payback time? [1]

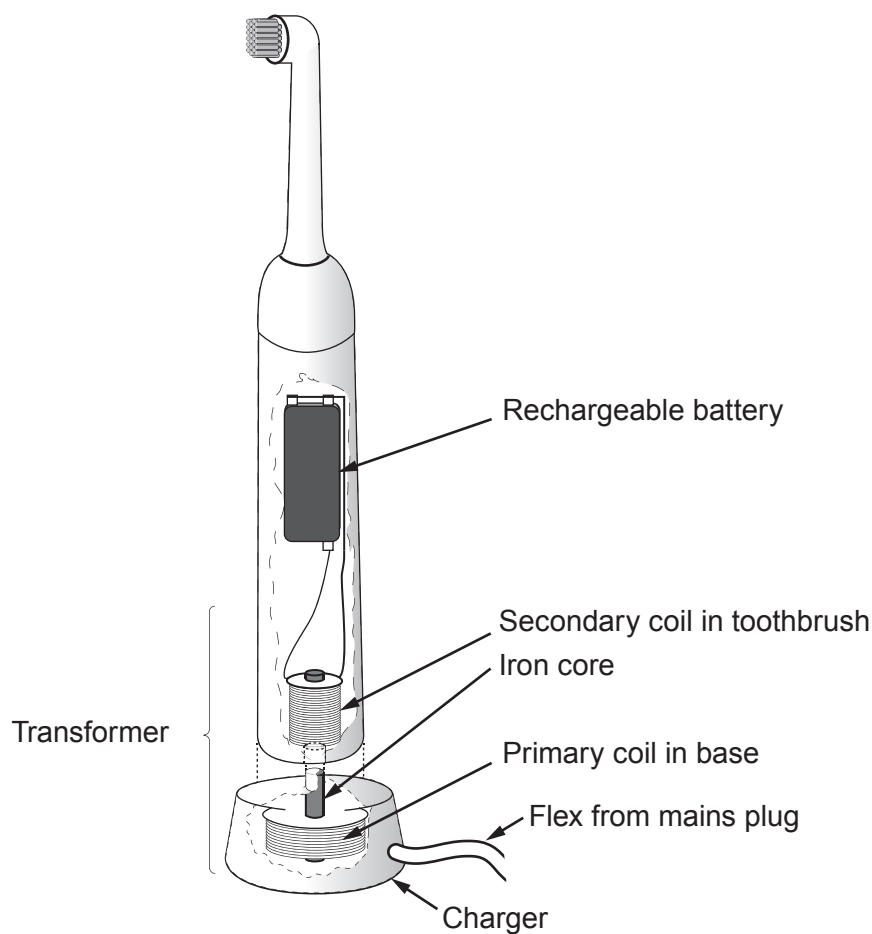
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3. The diagram shows a rechargeable toothbrush which contains a transformer.



(a) Explain how a voltage is produced in the secondary coil of the transformer. [3]

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(b) For the following questions, assume that the transformer is 100% efficient.

(i) Describe what is meant by *100% efficient*. [2]

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- (ii) The secondary coil inside the toothbrush has 480 turns. The primary coil in the base has 9 200 turns and is connected to the 230 V mains. Use an equation from page 2 to calculate the voltage across the secondary coil. [2]

Voltage = ..... V

- (iii) Given that the current produced in the secondary coil is 100 mA, use an equation from page 2 to calculate the power of the toothbrush. [3]

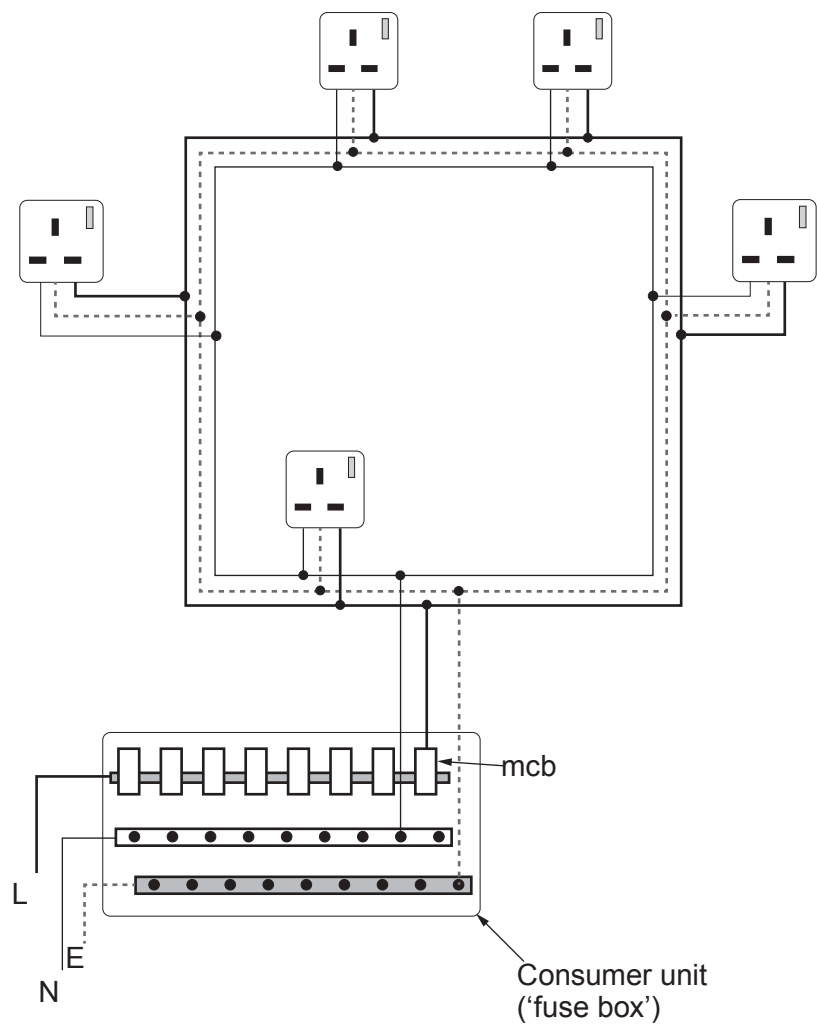
Power = ..... W

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4. (a) The diagram shows a simple ring main circuit for the kitchen of a house. The sockets are used to supply a tumble dryer with a current of 9 A, a kettle with 10 A and a toaster with 5 A all at the same time.



- (i) Describe the function of the earth wire. [2]

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- (ii) State **two** advantages of a miniature circuit breaker (mcb) over a fuse. [1]

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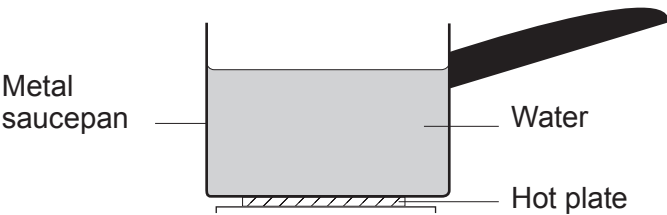
(iii) The cable of the ring main can safely have 21 A passing through it. Explain why it is safe to use all of these appliances at the same time. [2]

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(b) The diagram shows some water being heated in a metal saucepan.



Explain in terms of particles how heat is transferred to **all** of the water by conduction and convection. [4]

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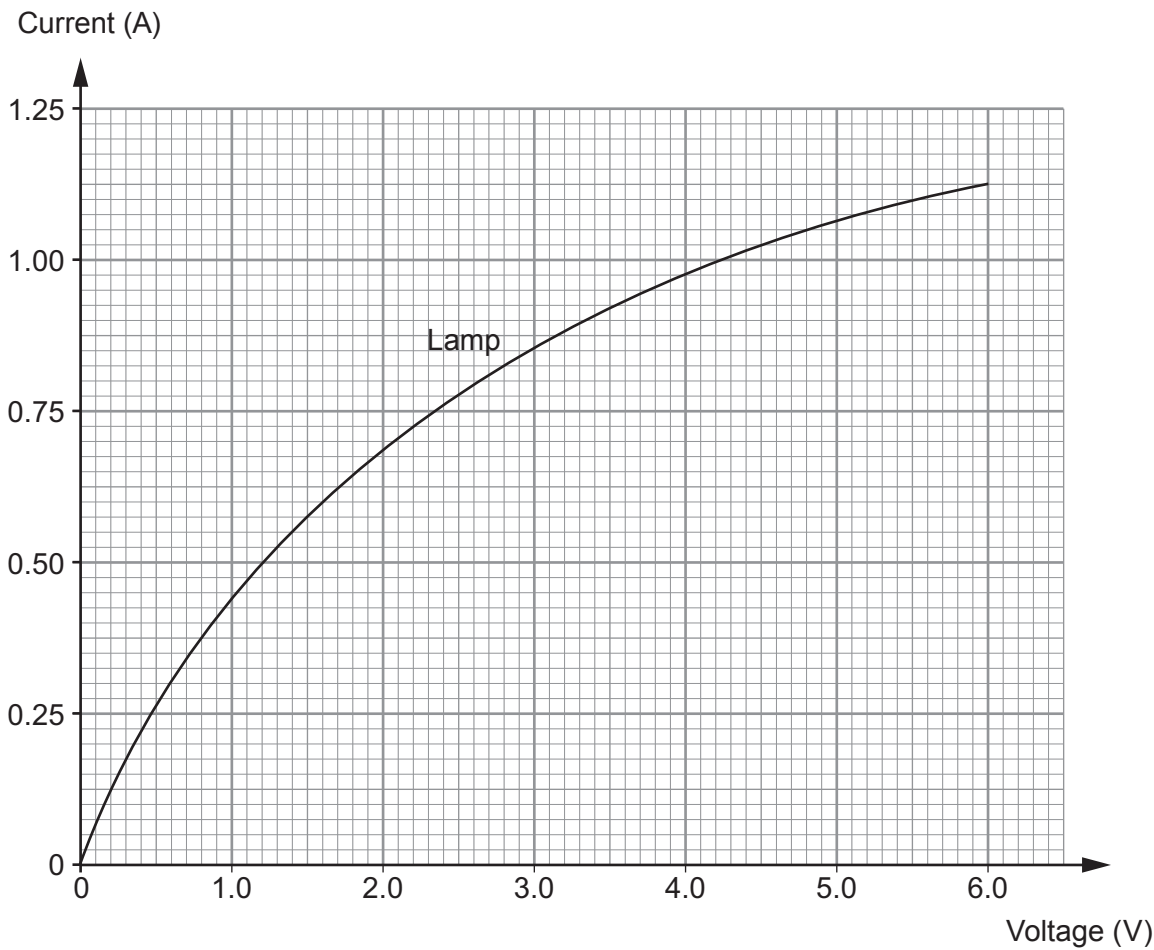
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5. In a practical lesson, a group of pupils investigate the voltage across and current through a lamp and a resistor. They draw a graph from their results for the **lamp**. It is shown below.

The current through the resistor is measured and recorded in a table which is shown below.

|                                  |      |      |      |      |
|----------------------------------|------|------|------|------|
| Voltage across the resistor (V)  | 1.0  | 2.0  | 4.0  | 6.0  |
| Current through the resistor (A) | 0.20 | 0.40 | 0.80 | 1.20 |

(a) (i) Plot the data for the resistor on the grid below and draw a suitable line. [2]



(ii) Describe how the current through the **lamp** changes as the voltage applied to it is varied. [2]

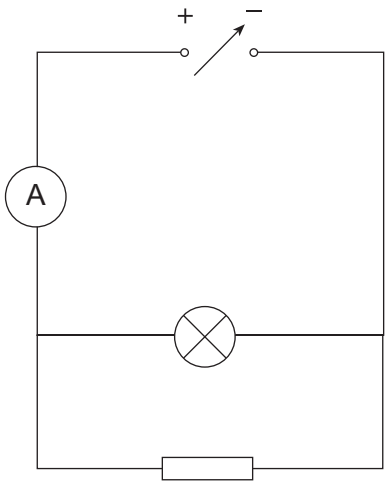
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- (b) Finally, the group connects the two components in parallel with a variable voltage power supply and an ammeter.



- (i) Use the graph and an equation from page 2 to calculate the resistance of the lamp at 3.0 V. [2]

Resistance = .....  $\Omega$

- (ii) Calculate the ammeter reading at 3.0 V. [2]

Current = ..... A

- (iii) Calculate the total resistance of the circuit at 3.0 V. [2]

Resistance = .....  $\Omega$

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- (iv) Explain, in terms of current, why the whole circuit has less resistance than the resistance of either component. [2]

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- (v) The voltage supply is increased from 0V to 6.0V. Explain how **the graph shows** that the lamp has a bigger resistance than the resistor at 6.0V. [2]

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- (vi) Use the graph and without calculations but giving clear reasoning, compare the powers of the lamp and the resistor between 0 and 6.0V. [2]

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6. (a) The meaning of the word “latent” is “hidden”. In terms of latent heat it suggests that the heat does something that is hidden.  
The value of the specific latent heat of fusion for ice is 334 000 J/kg and the value of the specific latent heat of vaporisation for water is 2 260 000 J/kg.  
Suggest why the word “latent” is used and account for the difference in the two values in terms of the behaviour of molecules. [6 QER]

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- (b) (i) A recycling company aims to melt 500 kg of aluminium in one hour. The aluminium, at an initial temperature of 20 °C is to be completely melted at 660 °C in a furnace of power 87 kW.

Use equations from page 2 and the information below to determine whether it is possible to meet the deadline. [6]

Specific heat capacity of aluminium,  $c = 900 \text{ J/kg } ^\circ\text{C}$   
Melting point of aluminium = 660 °C  
Specific latent heat of fusion of aluminium,  $L = 400\,000 \text{ J/kg}$

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- (ii) State why, in practice, it would take a lot longer than the time calculated above. [1]

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7. (a) (i) **Complete** the list showing the parts of the electromagnetic spectrum in order. [2]

gamma rays

X-rays

.....

visible light

infra-red

.....

radio waves

- (ii) State **one** property that is common to all electromagnetic waves. [1]

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- (b) Visible light has been used to accurately measure the distance between Earth and the Moon. In 1969, Neil Armstrong and Buzz Aldrin became the first humans to stand on the Moon. One of the jobs that they were required to do was to set up a reflector that faced Earth to allow light to be sent from the Earth, directed at it and be received back at Earth.

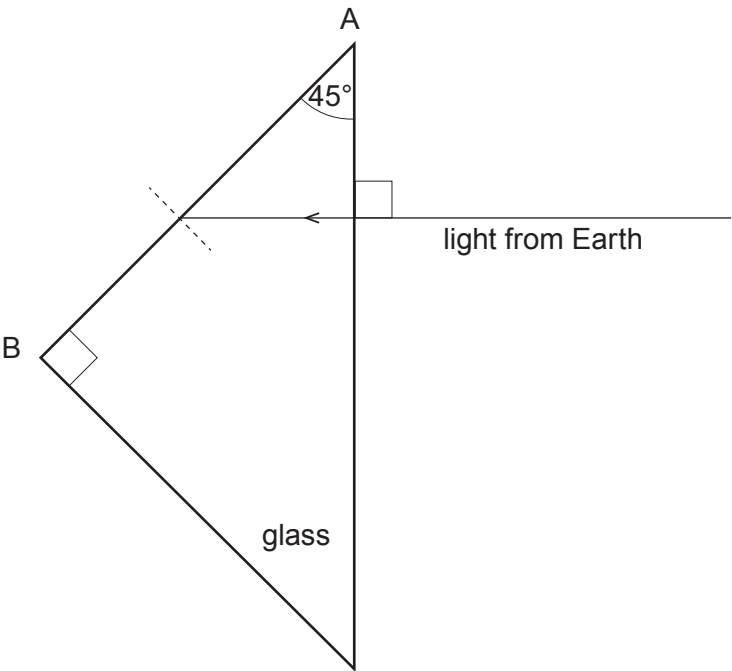
The reflectors have been used to accurately measure the distance from the Moon to Earth and the outcome is that the Moon is found to be getting further away from Earth by about 3.8 cm per year.



Laser light being sent  
to the Moon from a  
telescope in Germany.

The reflectors were illuminated from Earth by laser light of wavelength 532 nm. The reflectors are made from glass or plastic prisms. The light is sent in pulses separated by short intervals of time from a laser in a telescope on Earth.

The diagram shows a ray of light entering one of the prisms. The critical angle for the glass is  $36^\circ$ .



- (i) **Complete the diagram** to show the path followed by the light ray shown. [2]
- (ii) Explain why the ray of light would not be reflected back to Earth if it strikes the side AB of the prism at an angle of incidence of  $30^\circ$ . [2]

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- (iii) The wavelength of light used for measuring the distance to the Moon is  $532 \times 10^{-9}$  m and its frequency is  $5.639 \times 10^{14}$  Hz. Use equations from page 2 to calculate the distance between the Earth and the Moon if laser light is directed to the reflector and is received back again on Earth 2.5626 s later. [4]  
[All workings must be shown for each stage of your calculation.]

Distance = ..... m

- (iv) State why the laser light needs to be sent to the Moon in pulses separated by at least 2.57 s instead of in a continuous beam. [1]

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END OF PAPER

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| Surname     | Centre Number | Candidate Number |
| Other Names |               | 0                |



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**PHYSICS – Unit 1:**  
**Electricity, Energy and Waves**

**HIGHER TIER**

FRIDAY, 15 JUNE 2018 – MORNING

1 hour 45 minutes

| For Examiner’s use only |              |              |
|-------------------------|--------------|--------------|
| Question                | Maximum Mark | Mark Awarded |
| 1.                      | 15           |              |
| 2.                      | 11           |              |
| 3.                      | 8            |              |
| 4.                      | 9            |              |
| 5.                      | 9            |              |
| 6.                      | 13           |              |
| 7.                      | 15           |              |
| Total                   | 80           |              |

**ADDITIONAL MATERIALS**

In addition to this paper you will require a calculator, a ruler and a drawing compass..

**INSTRUCTIONS TO CANDIDATES**

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.  
Write your name, centre number and candidate number in the spaces at the top of this page.  
Answer **all** questions.  
Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet.

**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.  
The assessment of the quality of extended response (QER) will take place in question **6(a)**.



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## Equations

|                                                                                                                                                                                      |                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| current = $\frac{\text{voltage}}{\text{resistance}}$                                                                                                                                 | $I = \frac{V}{R}$                              |
| total resistance in a series circuit                                                                                                                                                 | $R = R_1 + R_2$                                |
| total resistance in a parallel circuit                                                                                                                                               | $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$  |
| energy transferred = power $\times$ time                                                                                                                                             | $E = Pt$                                       |
| power = voltage $\times$ current                                                                                                                                                     | $P = VI$                                       |
| power = current <sup>2</sup> $\times$ resistance                                                                                                                                     | $P = I^2R$                                     |
| % efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$                                                            |                                                |
| density = $\frac{\text{mass}}{\text{volume}}$                                                                                                                                        | $\rho = \frac{m}{V}$                           |
| units used (kWh) = power (kW) $\times$ time (h)<br>cost = units used $\times$ cost per unit                                                                                          |                                                |
| wave speed = wavelength $\times$ frequency                                                                                                                                           | $v = \lambda f$                                |
| speed = $\frac{\text{distance}}{\text{time}}$                                                                                                                                        |                                                |
| pressure = $\frac{\text{force}}{\text{area}}$                                                                                                                                        | $p = \frac{F}{A}$                              |
| $p$ = pressure<br>$V$ = volume<br>$T$ = kelvin temperature                                                                                                                           | $\frac{pV}{T} = \text{constant}$               |
|                                                                                                                                                                                      | $T / \text{K} = \theta / ^\circ\text{C} + 273$ |
| change in thermal energy = mass $\times$ specific heat capacity $\times$ change in temperature                                                                                       | $\Delta Q = mc\Delta\theta$                    |
| thermal energy for a change of state = mass $\times$ specific latent heat                                                                                                            | $Q = mL$                                       |
| force on a conductor (at right angles to a magnetic field) carrying a current = magnetic field strength $\times$ current $\times$ length                                             | $F = BIl$                                      |
| $V_1$ = voltage across the primary coil<br>$V_2$ = voltage across the secondary coil<br>$N_1$ = number of turns on the primary coil<br>$N_2$ = number of turns on the secondary coil | $\frac{V_1}{V_2} = \frac{N_1}{N_2}$            |

## SI multipliers

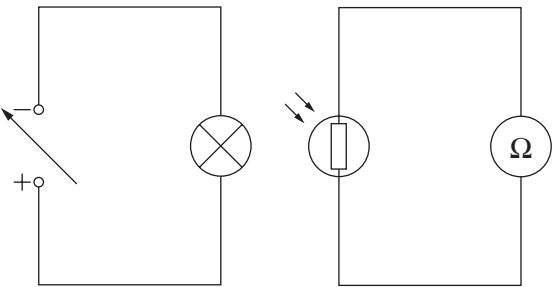
| Prefix | Multiplier          |
|--------|---------------------|
| p      | $1 \times 10^{-12}$ |
| n      | $1 \times 10^{-9}$  |
| $\mu$  | $1 \times 10^{-6}$  |
| m      | $1 \times 10^{-3}$  |

| Prefix | Multiplier         |
|--------|--------------------|
| k      | $1 \times 10^3$    |
| M      | $1 \times 10^6$    |
| G      | $1 \times 10^9$    |
| T      | $1 \times 10^{12}$ |



Answer all questions.

1. The following circuits are set up to investigate a light dependent resistor (LDR). The voltage of the power supply is changed to vary the power of the lamp to alter its brightness. The resistance of the LDR is measured with an ohmmeter ( $\Omega$ ) for each power of the lamp.



- (a) (i) State **two** variables, **other than using the same components**, that should be controlled in this experiment. [2]
1. ....
2. ....
- (ii) Explain how the design of the experiment could be improved to make the results more valid. [2]

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- (b) The results are shown in the table below.

| Power of lamp (W) | Resistance of LDR (k $\Omega$ ) |
|-------------------|---------------------------------|
| 2                 | 19.5                            |
| 4                 | 10.3                            |
| 8                 | 3.0                             |
| 12                | 2.2                             |
| 16                | 1.5                             |
| 20                | 1.3                             |
| 24                | 1.1                             |

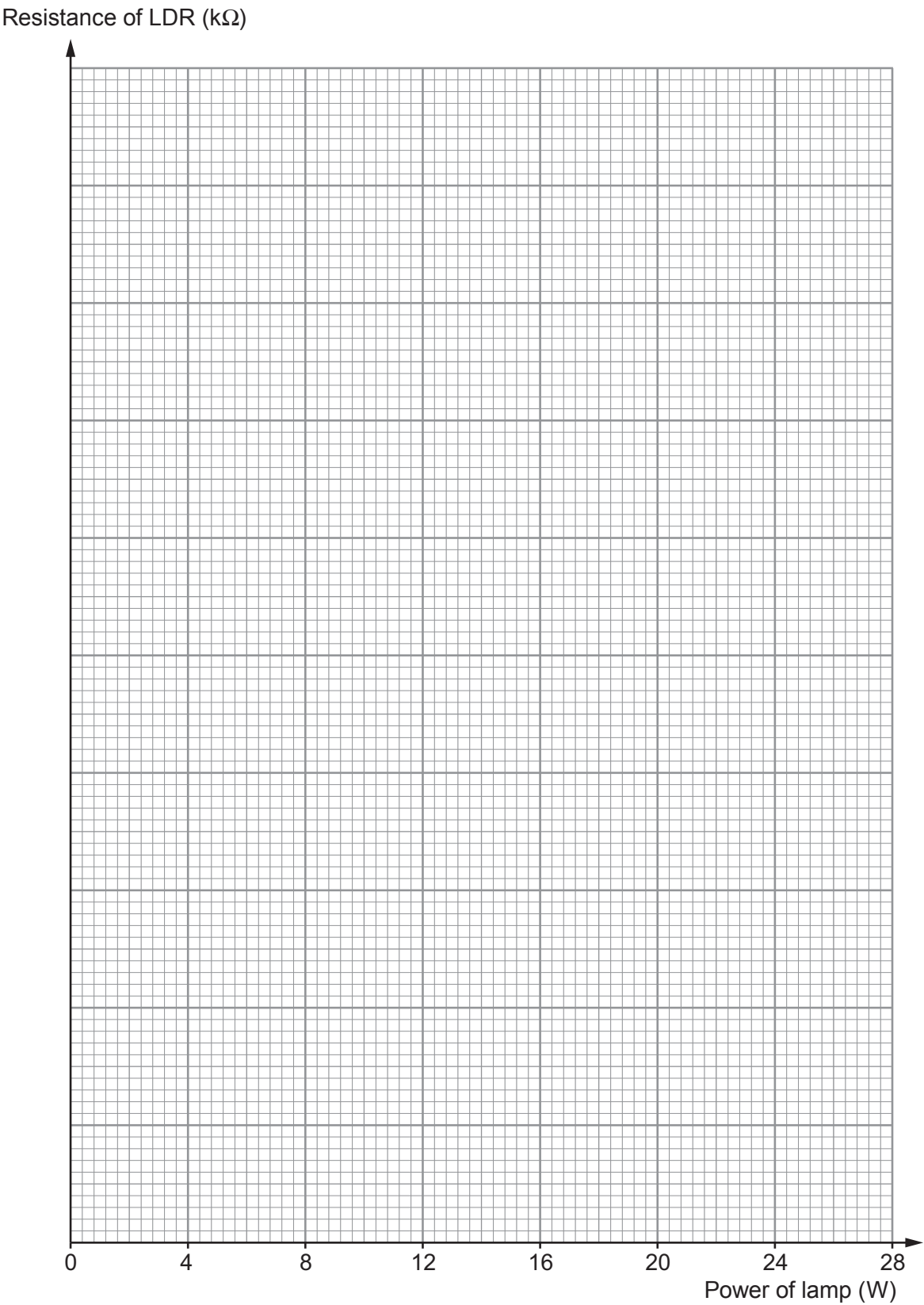




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(i) Use the data to plot a graph on the grid below and draw a suitable line.

[4]



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(ii) Use the graph to find the resistance of the LDR for a lamp power of 10 W. [1]

Resistance = .....  $\Omega$

(iii) It is suggested that when the lamp power doubles, the LDR resistance halves. Explain, using values from the table, to what extent this suggestion is true. [3]

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(c) The LDR is connected into another circuit. The voltage across the LDR is 2.8 V and the current through it is 0.35 mA. Use an equation from page 2 to calculate the resistance of the LDR. [3]

Resistance = .....  $\Omega$

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2. The **epicentre** is the point on the Earth’s surface directly above an earthquake. Seismic stations detect earthquakes by the tracings made on seismographs.

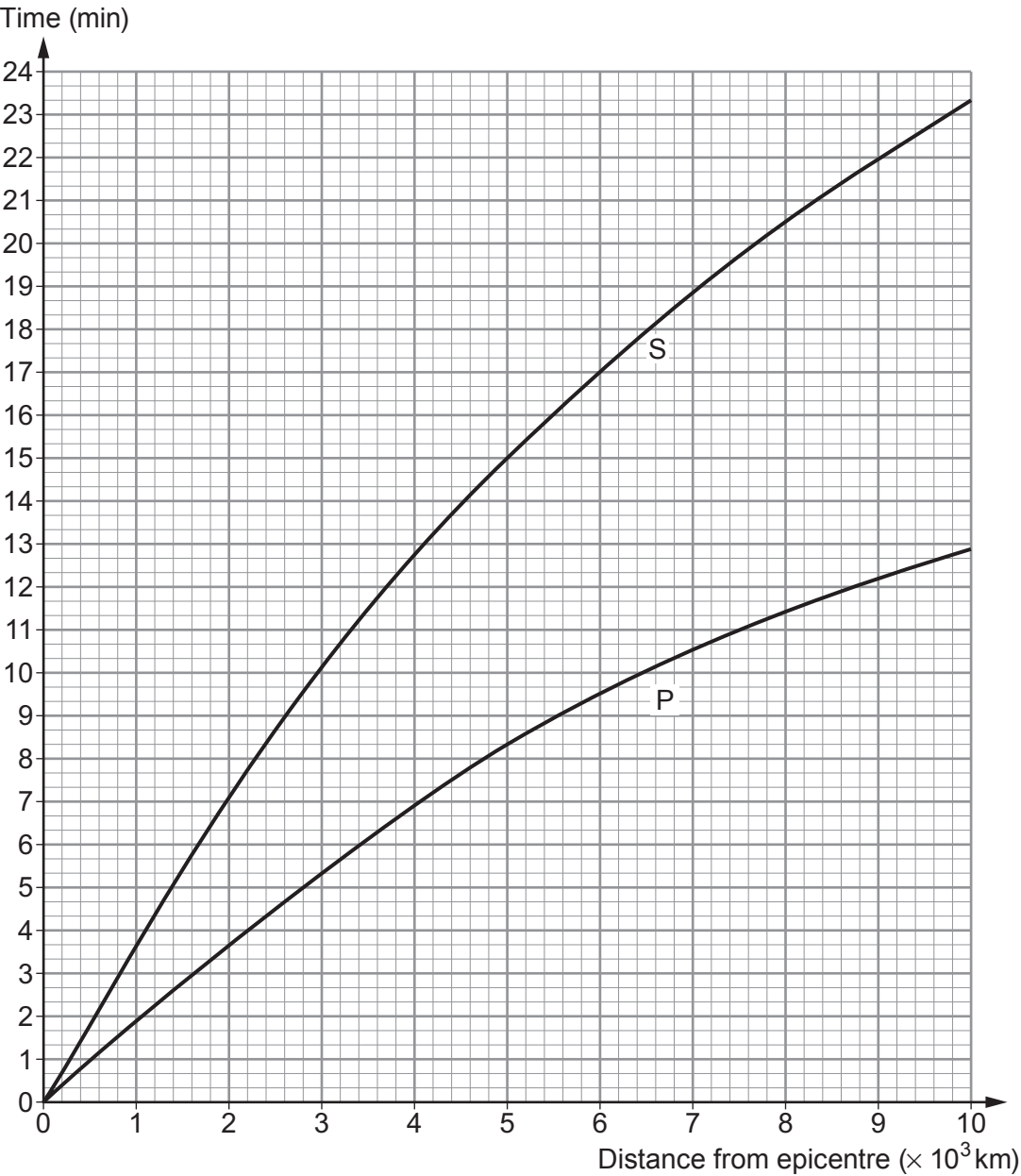
(a) Surface, P and S waves are three types of earthquake waves.  
Tick (✓) the boxes next to the **three** correct statements about earthquake waves. [3]

- Surface waves travel the fastest ☐
- S waves travel on the surface of the Earth ☐
- S waves are transverse waves ☐
- P waves travel through solids and liquids ☐
- P waves are longitudinal waves ☐
- S waves cause the most damage ☐

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(b) The graph shows the time taken by P and S waves to travel different distances from the epicentre.



Each small square on the time axis represents 20 s.

(i) Use the graph to answer the following questions.

I. State the time it takes for a P wave to travel  $5 \times 10^3$  km from the epicentre. [1]

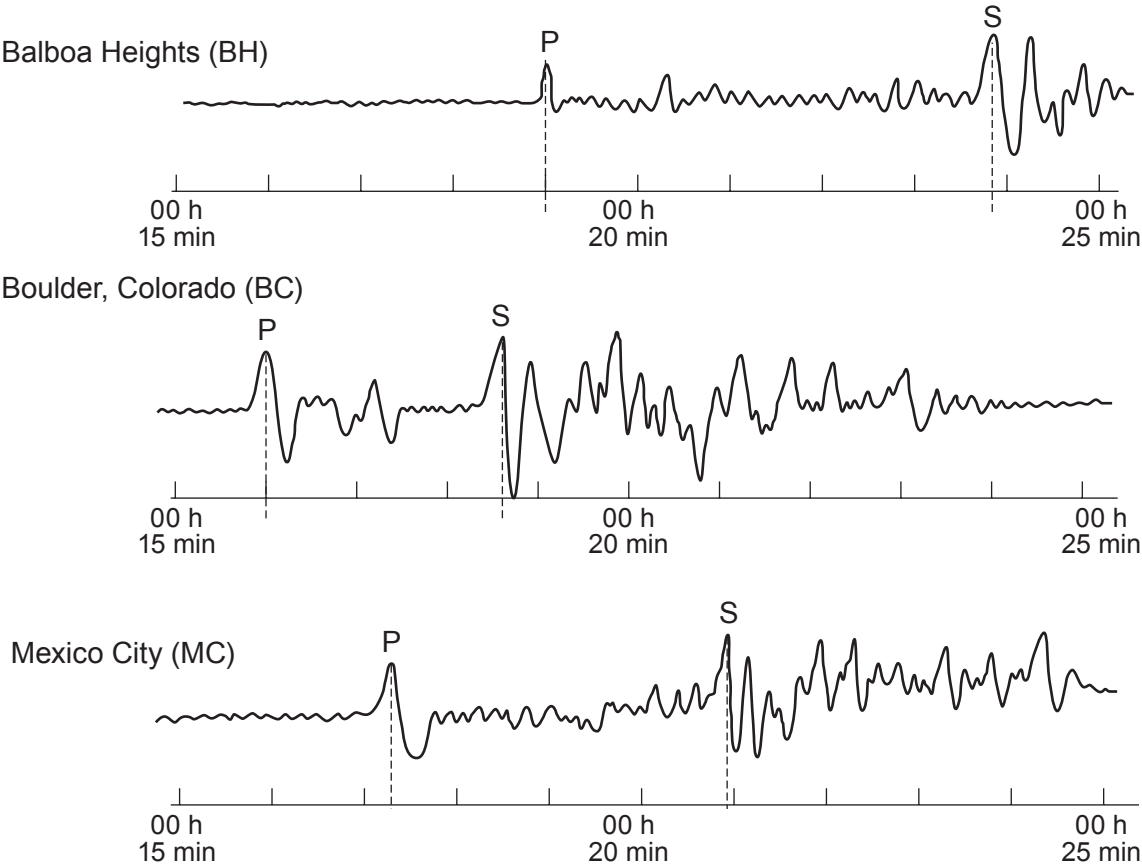
time = ..... min ..... s

II. State the **extra** time it takes S waves to travel  $5 \times 10^3$  km from the epicentre. [1]

time = ..... min ..... s



(ii) Study the three seismograph tracings below. Tracings made at three separate seismic stations are needed to locate an earthquake epicentre. P shows the arrival of P waves and S shows the arrival of S waves.



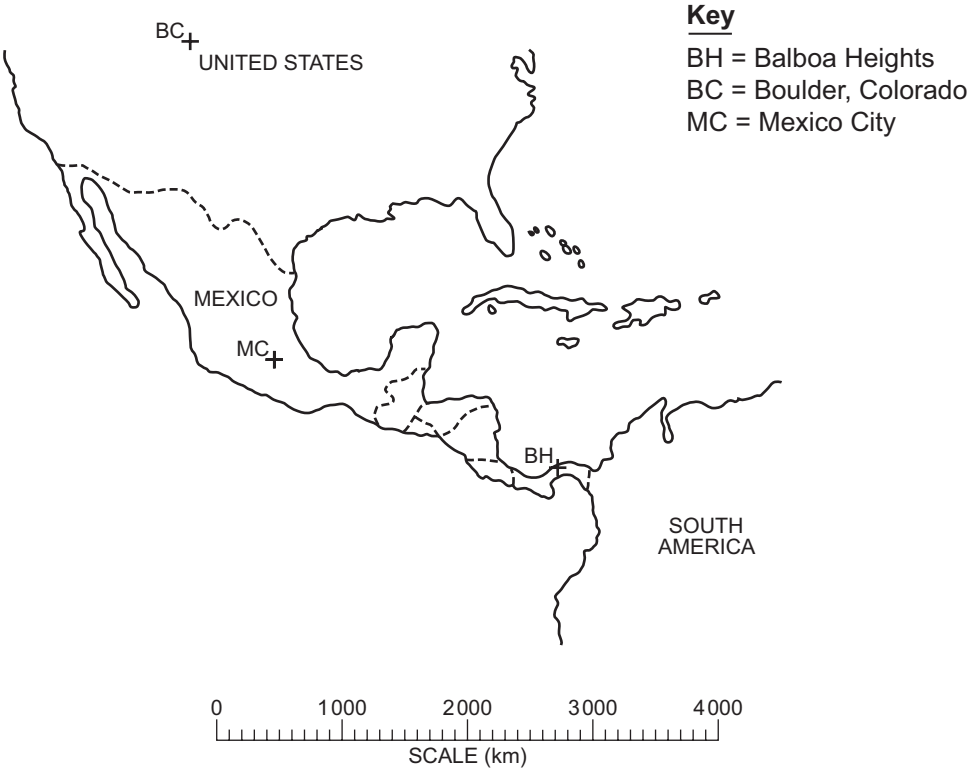
Use the information in the graph and tracings to **complete the table**. [3]

| City                   | Arrival time of P waves<br>(h:min:s) | Arrival time of S waves<br>(h:min:s) | Time difference for P and S waves<br>(h:min:s) | Distance to epicentre<br>( $\times 10^3$ km) |
|------------------------|--------------------------------------|--------------------------------------|------------------------------------------------|----------------------------------------------|
| Balboa Heights (BH)    | 00:19:00                             | 00:23:50                             | 00:04:50                                       | 3.2                                          |
| Boulder, Colorado (BC) | ..... : ..... : .....                | 00:18:40                             | ..... : ..... : .....                          | .....                                        |
| Mexico City (MC)       | 00:17:15                             | 00:20:55                             | 00:03:40                                       | 2.2                                          |



Examiner  
only

(iii) Use the data in the table to find the epicentre of the earthquake on the diagram below. [3]



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Examiner  
only

3. (a) Measurements from an experiment to find the density of an irregular shaped solid are given below.

Mass of the solid = 26.0 g  
Volume of water in measuring cylinder = 40 cm<sup>3</sup>  
Volume of water and solid = 48 cm<sup>3</sup>

Use an equation from page 2 and the information above to calculate the density of the solid. [3]

Density = ..... g/cm<sup>3</sup>

- (b) Describe how you would change the experiment to find the density of an irregular shaped solid that floated on water. [3]

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- (c) Explain in terms of particles why most solids have greater densities than water. [2]

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Examiner  
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4. (a) Describe the advantages of household ring main circuits. [3]

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(b) Explain which safety device operates when the following faults occur.

(i) The ring main circuit stops working when an additional fault-free appliance is plugged into it. [2]

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(ii) The circuit stops working when a lawnmower accidentally cuts the power cable during use. [2]

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(c) Explain why electrical appliances with a metal casing require an earth lead whereas those with a plastic casing do not. [2]

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Examiner  
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5. Satellites are used to communicate between base stations at different places on Earth.

- (a) (i) **Complete the diagram** below to show how base station A communicates with base station B. [3]



- (ii) Explain why a geostationary satellite must be used for constant communication rather than a geosynchronous one. [2]

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- (b) A base station communicates with another base station using a single geostationary satellite  $3.6 \times 10^4$  km above the Earth. They communicate with each other using microwaves of wavelength 2.8 cm travelling at  $3 \times 10^8$  m/s.

Use an equation from page 2 to calculate the time delay between sending a signal from one base station and receiving the signal at the other. [4]

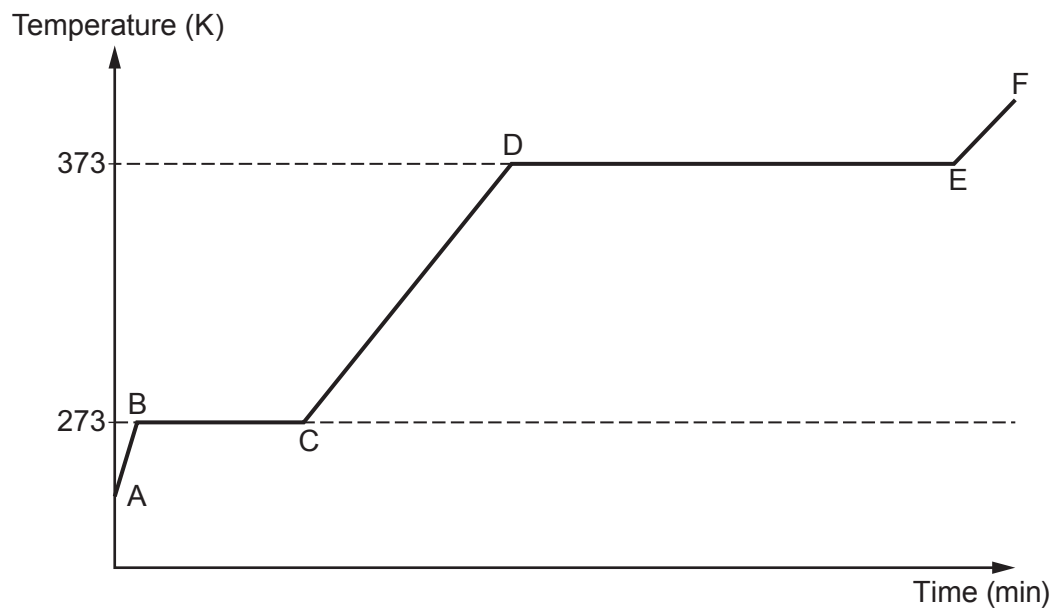
Time = ..... s

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Examiner  
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6. The graph below shows how the temperature of a block of ice changes over time as it is heated.



- (a) Refer to the graph to explain the changes in temperature and changes in state of the block of ice in terms of the behaviour of molecules. [6 QER]



Examiner  
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(b) The specific latent heat,  $L$ , of fusion of ice is 336 000 J/kg.

(i) Explain what is meant by this statement. [2]

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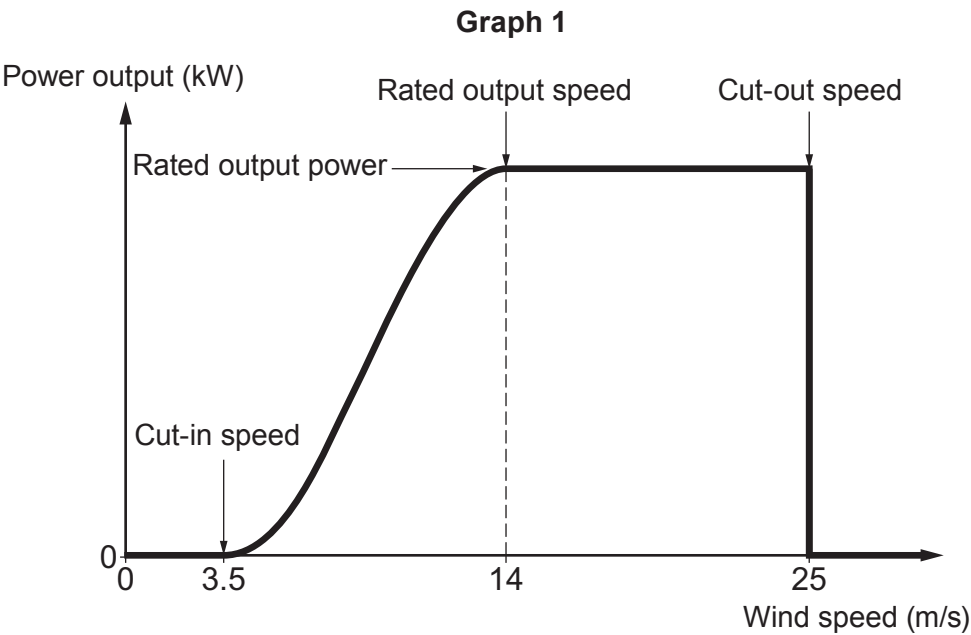
(ii) The ice block has a mass of 800 g. Its initial temperature is 250 K.  
Use equations from page 2 to calculate the energy required to raise its temperature to 273 K and completely melt the ice block at 273 K.  
The specific heat capacity,  $c$ , of ice is 2030 J/kg K. [5]

Energy = ..... J

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7. Wind turbines are mounted on towers to capture the most energy. They harness the wind’s energy with their propeller-like blades. Wind turbines can be used as stand-alone applications, or they can be connected to the National Grid. For larger scale sources of wind energy, a number of wind turbines are built close together to form a wind farm. Several electricity providers today use wind farms to supply power to their customers.
- (a) The power output from a wind turbine is not constant but depends on wind speed as shown in Graph 1.

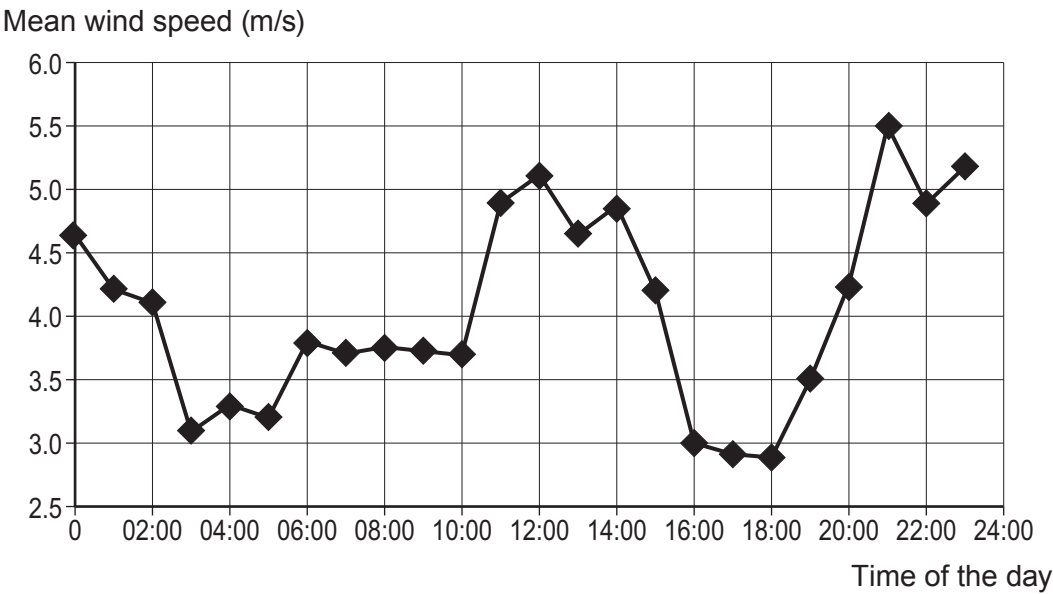


The cut-in speed is the speed at which the turbine first starts to rotate and generate power. Somewhere between 12 and 15 m/s, the output power reaches the maximum limit. This limit is called the rated output power and the wind speed at which it is reached is called the rated output speed. As the speed increases the forces on the turbine structure continue to rise and, at some point, there is a risk of damage to the rotor. As a result, a braking system is employed to bring the rotor to a standstill. This occurs at the cut-out speed and is usually around 25 m/s.



The mean wind speed for different times during one day in May are shown in Graph 2.

Graph 2



It is claimed that wind power is **always** a useful back up at times of high demand between the hours of 06:00 to 10:00 and between 16:00 to 20:00 because it is always windy.

Explain whether the information in both Graphs 1 and 2 support this claim. [4]

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TURN OVER FOR THE REST OF THE QUESTION



Examiner  
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- (b) A wind farm off the coast of Wales, has an output voltage of 16 kV and a total rated output power of 24 MW. It is connected to the National Grid by a transformer. The transformer has an input coil containing 400 turns and an output coil containing 4 800 turns. Assume the transformer has an efficiency of 100%.

Use equations from page 2 to answer the following questions.

- (i) Calculate the output voltage of the transformer in kV. [2]

Output voltage = ..... kV

- (ii) Calculate the output current of the transformer. [3]

Output current = ..... A

- (c) (i) Explain the role of transformers in the National Grid. [3]

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(ii) Explain the purpose of an iron core in a transformer **and** state why it is laminated. [3]

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Examiner  
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END OF PAPER



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| Surname     | Centre Number | Candidate Number |
| Other Names |               | 0                |



GCSE  
3420UA0-1



FRIDAY, 14 JUNE 2019 – MORNING

PHYSICS – Unit 1:  
Electricity, Energy and Waves

HIGHER TIER  
1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.  
Write your name, centre number and candidate number in the spaces at the top of this page.  
Answer **all** questions.  
Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.  
The assessment of the quality of extended response (QER) will take place in question **6(a)**.



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## Equations

|                                                                                                                                                                                      |                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| current = $\frac{\text{voltage}}{\text{resistance}}$                                                                                                                                 | $I = \frac{V}{R}$                              |
| total resistance in a series circuit                                                                                                                                                 | $R = R_1 + R_2$                                |
| total resistance in a parallel circuit                                                                                                                                               | $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$  |
| energy transferred = power $\times$ time                                                                                                                                             | $E = Pt$                                       |
| power = voltage $\times$ current                                                                                                                                                     | $P = VI$                                       |
| power = current <sup>2</sup> $\times$ resistance                                                                                                                                     | $P = I^2R$                                     |
| % efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$                                                            |                                                |
| density = $\frac{\text{mass}}{\text{volume}}$                                                                                                                                        | $\rho = \frac{m}{V}$                           |
| units used (kWh) = power (kW) $\times$ time (h)<br>cost = units used $\times$ cost per unit                                                                                          |                                                |
| wave speed = wavelength $\times$ frequency                                                                                                                                           | $v = \lambda f$                                |
| speed = $\frac{\text{distance}}{\text{time}}$                                                                                                                                        |                                                |
| pressure = $\frac{\text{force}}{\text{area}}$                                                                                                                                        | $p = \frac{F}{A}$                              |
| $p$ = pressure<br>$V$ = volume<br>$T$ = kelvin temperature                                                                                                                           | $\frac{pV}{T} = \text{constant}$               |
|                                                                                                                                                                                      | $T / \text{K} = \theta / ^\circ\text{C} + 273$ |
| change in thermal energy = mass $\times$ specific heat capacity $\times$ change in temperature                                                                                       | $\Delta Q = mc\Delta\theta$                    |
| thermal energy for a change of state = mass $\times$ specific latent heat                                                                                                            | $Q = mL$                                       |
| force on a conductor (at right angles to a magnetic field) carrying a current = magnetic field strength $\times$ current $\times$ length                                             | $F = BIl$                                      |
| $V_1$ = voltage across the primary coil<br>$V_2$ = voltage across the secondary coil<br>$N_1$ = number of turns on the primary coil<br>$N_2$ = number of turns on the secondary coil | $\frac{V_1}{V_2} = \frac{N_1}{N_2}$            |

## SI multipliers

| Prefix | Multiplier          |
|--------|---------------------|
| p      | $1 \times 10^{-12}$ |
| n      | $1 \times 10^{-9}$  |
| $\mu$  | $1 \times 10^{-6}$  |
| m      | $1 \times 10^{-3}$  |

| Prefix | Multiplier         |
|--------|--------------------|
| k      | $1 \times 10^3$    |
| M      | $1 \times 10^6$    |
| G      | $1 \times 10^9$    |
| T      | $1 \times 10^{12}$ |



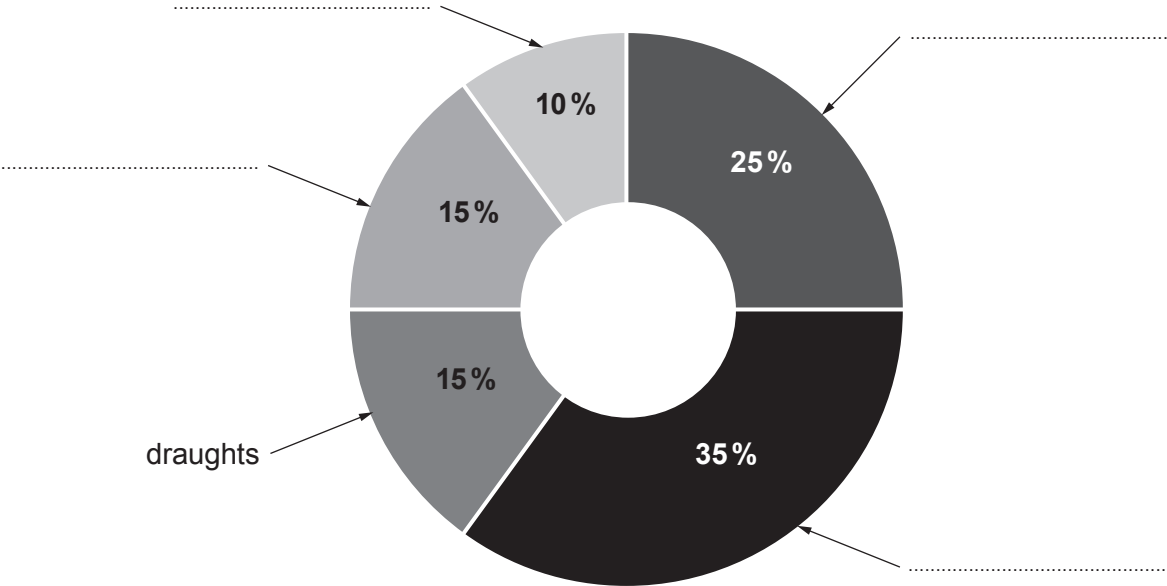
Answer all questions.

1. The Welsh Government Warm Homes Scheme, called Nest, aims to make Welsh homes warmer and more energy efficient places to live. Nest is accessible to all homeowners in Wales and provides advice on saving energy.

(a) Draughts, floors, windows, walls and the roof are the five ways energy is lost from a heated house. The pie chart shows the percentages of energy loss by each of the five ways.

Use the information below to **complete the labelling** on the pie chart. [3]  
One label has been completed already.

- The percentage energy loss through the walls is greatest.
- Windows and draughts have equal percentage energy losses.
- Windows and floor percentage losses add up to equal the percentage loss through the roof.



(b) To reduce energy loss through the roof the Nest Scheme suggests the installation of fibre-glass insulation in the loft.

(i) Explain how fibre-glass reduces energy loss in the loft by convection. [2]

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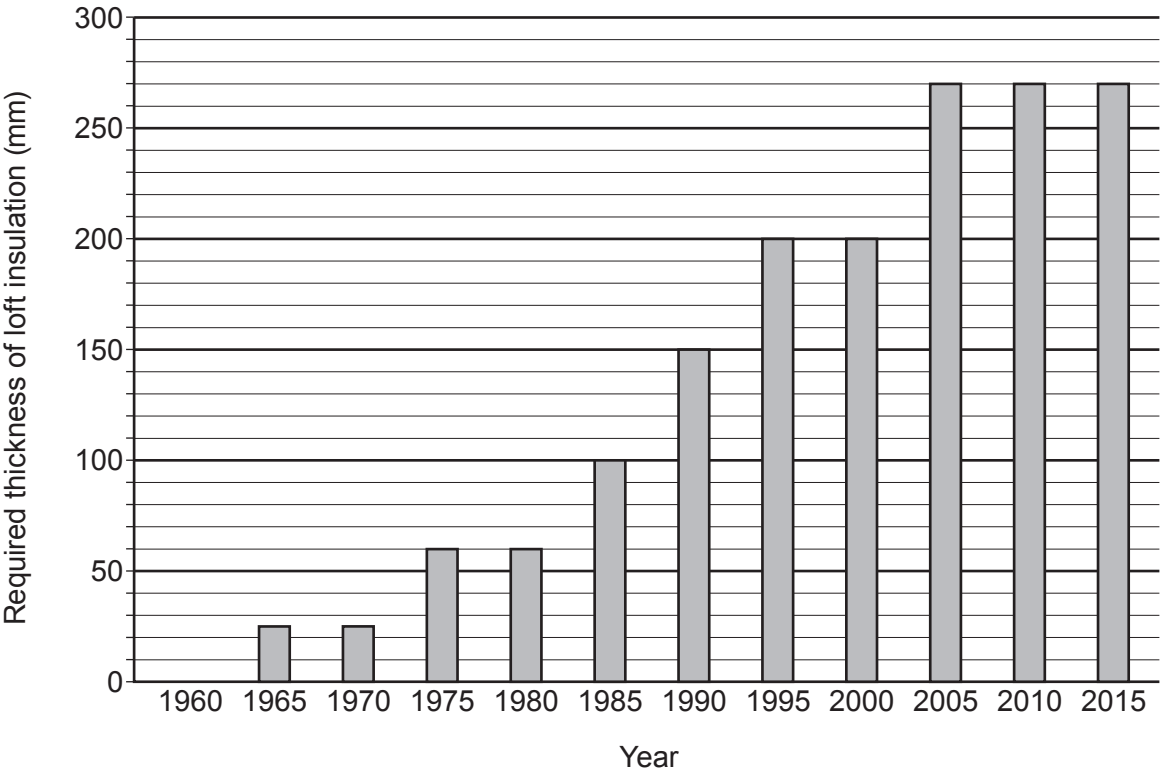
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- (ii) There are strict regulations about the minimum thickness of insulation that must be installed in the loft of a new house. The graph shows how the required thickness of loft insulation has changed since 1960.



Use information from the graph to tick (✓) the **three** correct statements. [3]

- Every five years the required thickness of loft insulation increases. ☐
- The required thickness of loft insulation in 2000 is 8 times thicker than in 1970. ☐
- In 1960 houses did not lose any energy through their roof. ☐
- A house built in 1980 needs 210 mm of loft insulation added to bring it up to 2015 standards. ☐
- The **general** trend of the graph indicates that the required thickness of loft insulation has increased at an increasing rate. ☐
- The required thickness of loft insulation will remain constant after 2015. ☐



Examiner  
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- (iii) A homeowner must install loft insulation in a new extension. **It has a loft area of 120 m<sup>2</sup>.** The insulation must be at least 270 mm thick to meet building regulations.

There is a selection of fibre-glass insulations available.

|                                          | Insulation 1<br>(270 mm thick) | Insulation 2<br>(350 mm thick) | Insulation 3<br>(300 mm thick) |
|------------------------------------------|--------------------------------|--------------------------------|--------------------------------|
| Installation cost<br>(£/m <sup>2</sup> ) | 3.50                           | 5.55                           | 4.50                           |
| Estimated saving<br>per year (£)         | 84                             | 111                            | 98                             |
| Payback time<br>(years)                  | 5.0                            | 6.0                            |                                |

Calculate the payback time if **insulation 3** was installed in the 120 m<sup>2</sup> extension. [2]

Payback time = ..... years

- (iv) The homeowner considers installing **insulation 1** as it is cheapest but the builder says that **insulation 2** should be installed as it will save more money over 40 years. Explain, with calculations, whether the builder is correct. [2]

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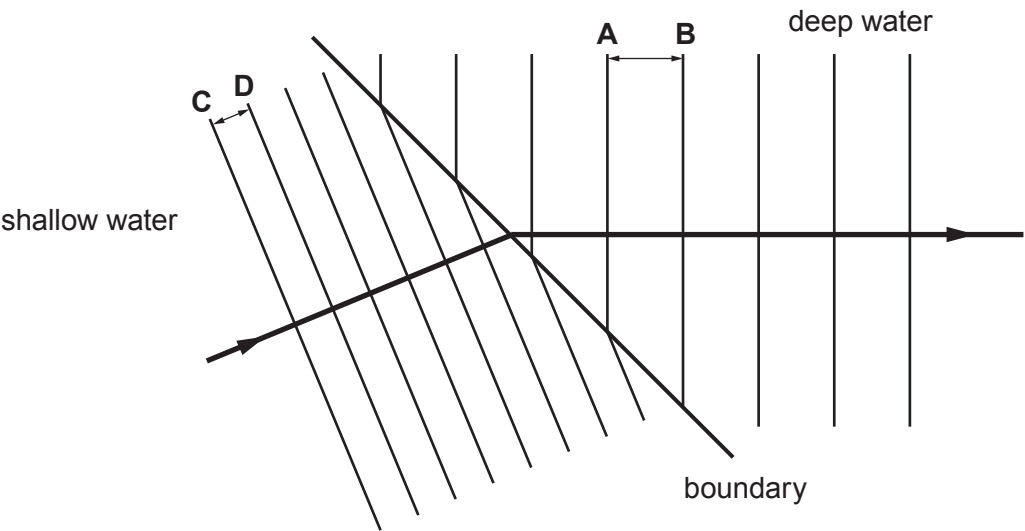
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2. In class, a teacher demonstrates refraction using a ripple tank. The diagram below shows plane wavefronts travelling across a boundary between shallow and deep water. The frequency of the waves remains **constant** during refraction.



- (a) Using a ruler, students measure the distance between wavefronts **A** and **B**. This measurement is the wavelength of the water waves in deep water. The distance between wavefronts **C** and **D** is measured to obtain their wavelength in the shallow water. The results are shown below.

|                 | Deep water ( <b>AB</b> ) | Shallow water ( <b>CD</b> ) |
|-----------------|--------------------------|-----------------------------|
| Wavelength (mm) | 10                       | 5                           |

- (i) State how the measurement of wavelength could be improved. [1]

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- (ii) The wavelength in the deep water is twice the wavelength in the shallow water. The teacher suggests, “the speed of the wavefronts in shallow water is double the speed of the wavefronts in the deep water.” Using information provided explain if the suggestion made by the teacher is correct. [2]

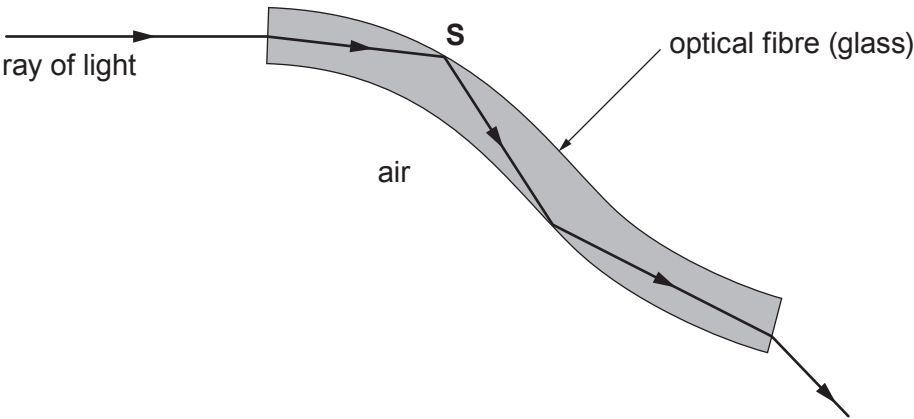
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- (b) An endoscope uses optical fibres. It can be used by doctors to produce medical images of a specific area inside a patient. A bundle of fibres is inserted into the body. Some of the fibres carry light into the body and others return the light reflected off internal surfaces. The diagram shows a ray of light passing through part of an optical fibre of an endoscope.



- (i) State the name given to the change in direction of the signal at **S**. [1]

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- (ii) State the **two** conditions needed for the ray of light to change direction at **S**. [2]

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- (iii) Medical images can also be obtained from a computer tomography (CT) scan. This type of scan uses X-rays targeted at the patient from different positions outside the body. The information collected is processed by a computer to produce detailed 3D image segments of the patient.  
Explain a disadvantage of using a CT scan to obtain medical information compared to using an endoscope. [2]

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Examiner  
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3. The diagram shows the different waves that make up the electromagnetic (em) spectrum.

|       |        |             |               |           |            |             |
|-------|--------|-------------|---------------|-----------|------------|-------------|
| Gamma | X-rays | Ultraviolet | Visible light | Infra-red | Microwaves | Radio waves |
|-------|--------|-------------|---------------|-----------|------------|-------------|

(a) State in terms of **two** of their properties why they are arranged in this order from left to right. [2]

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(b) (i) State which em wave can be radiated as energy from the nuclei of radioactive materials. [1]

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(ii) State which em wave is used for satellite communications. [1]

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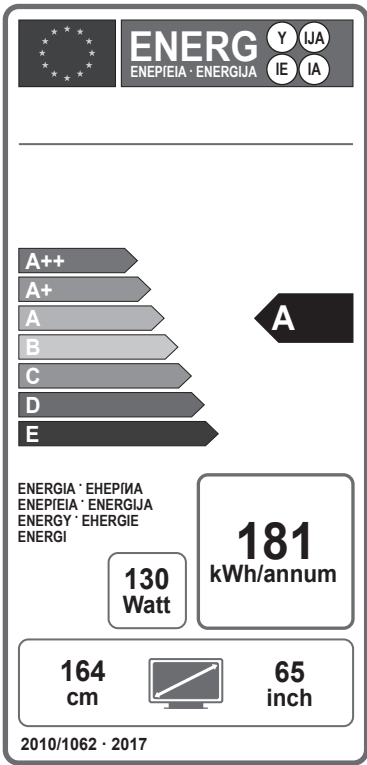
(c) The energy of X-rays ranges from  $2.0 \times 10^{-14} \text{ J}$  to  $2.1 \times 10^{-17} \text{ J}$ . Which em wave would have an energy of  $1.5 \times 10^{-12} \text{ J}$ ? [1]

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4. The following label is attached to a new TV.



(a) A few years ago the energy efficiency banding had a range from G up to A. Give a reason why additional coding, A+ and A++, has recently been added to the banding. [1]

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(b) The power of the TV is shown as 130 W. The TV manufacturer estimates that, on average, the TV will use 181 kWh per year. Using an equation from page 2, calculate the mean number of hours it is used per day. There are 365 days in a year. [4]

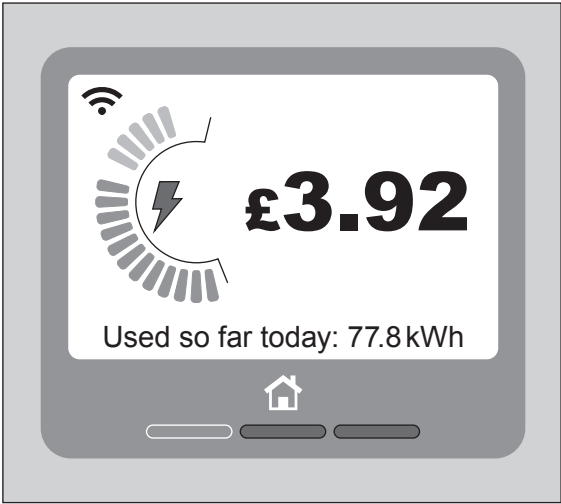
Mean number of hours used per day = .....





Examiner  
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- (c) Smart meters are a new kind of energy meter. As part of the government’s plan to update the U.K.’s energy system they would like all homes to have Smart meters fitted by the end of 2020. Here is a screenshot of the Smart meter display used on a promotional leaflet.



Use an equation from page 2 to calculate the cost **in pence** of each kWh based on this display. [2]

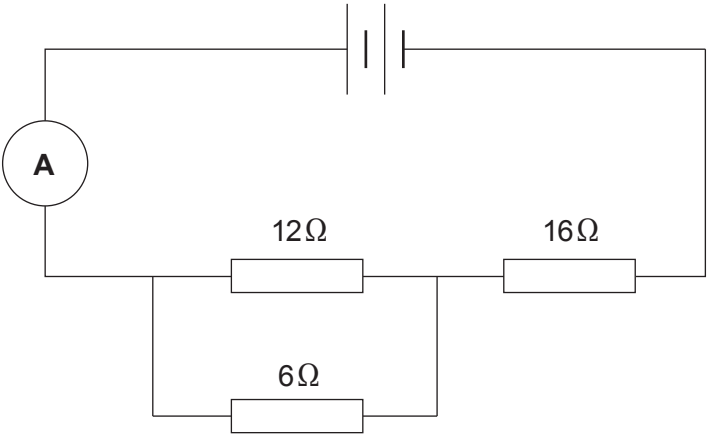
Cost of kWh = ..... p

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5. In an experiment a  $16\Omega$ ,  $12\Omega$  and  $6\Omega$  resistor are connected as shown.



(a) Use equations from page 2 to calculate the total resistance of the circuit. [3]

Total resistance = .....  $\Omega$

(b) A brand new battery is used in the circuit. The manufacturer states that the new battery can transfer a maximum of  $75.6\text{ kJ}$  of energy. The current from the battery is measured as  $1.5\text{ A}$ .

Use the equation:

$$P = I^2 R$$

and another equation from page 2, to calculate the time **in minutes** the battery will last in this circuit. [4]

Time = ..... minutes

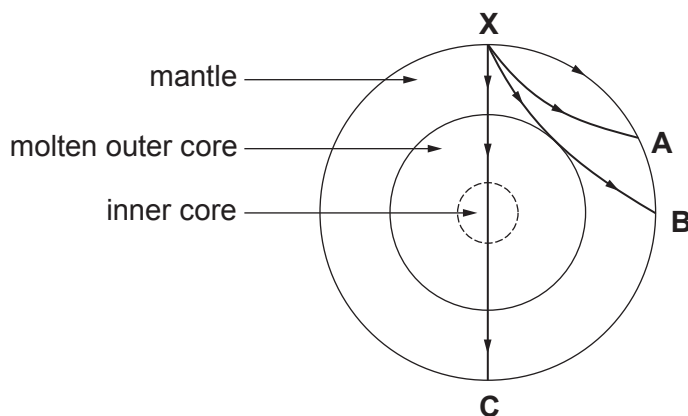
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Examiner  
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6. Earthquake monitoring stations can detect P and S waves from earthquakes. A seismic trace contains information about times of arrival for each wave type.

- (a) The diagram shows some paths taken by seismic waves travelling from an earthquake at **X**. There are three seismic monitoring stations: **A**, **B** and **C**.



Compare the seismic traces obtained at **A**, **B** and **C**, explaining any similarities **and** any differences. [6 QER]



(b) S waves are transverse waves and P waves are longitudinal waves. Describe the difference between transverse and longitudinal waves.

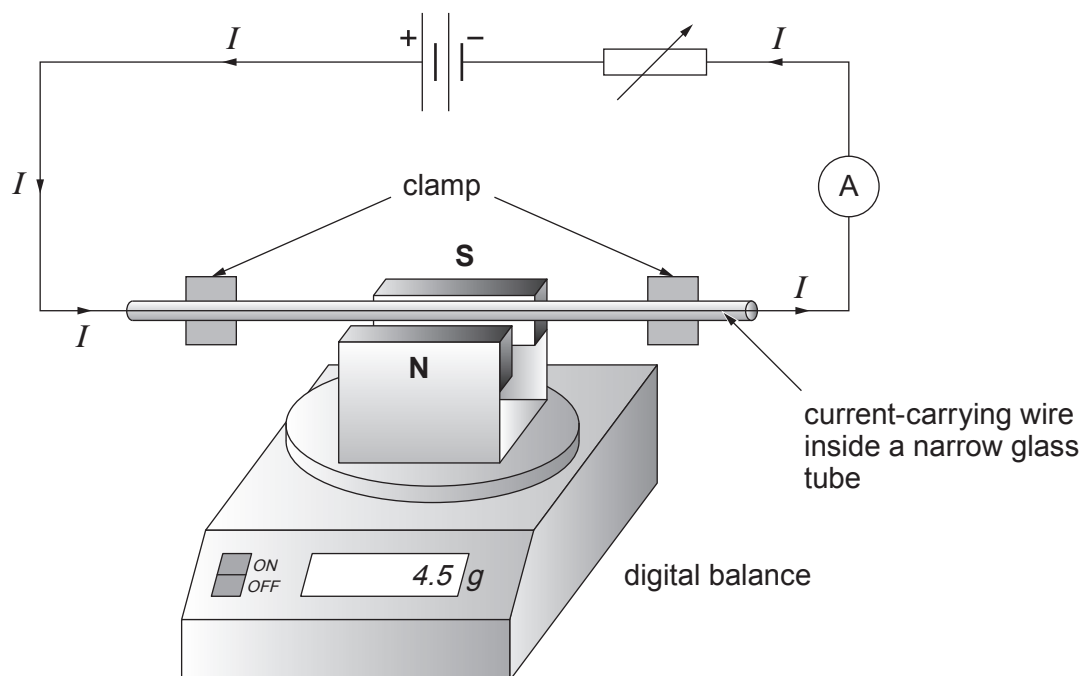
[2]

Examiner  
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7. Apparatus is set up in a laboratory to investigate the force exerted by a current-carrying wire when it is placed in a uniform magnetic field. A magnet is placed on a digital balance and a current-carrying wire is placed between its magnetic poles.



- (a) To prevent the current-carrying wire moving during the experiment it is contained in a narrow glass tube that is clamped. A pupil predicts that the current-carrying wire would move upwards if it had **not** been contained in the narrow glass tube. Explain whether the pupil is correct. [3]

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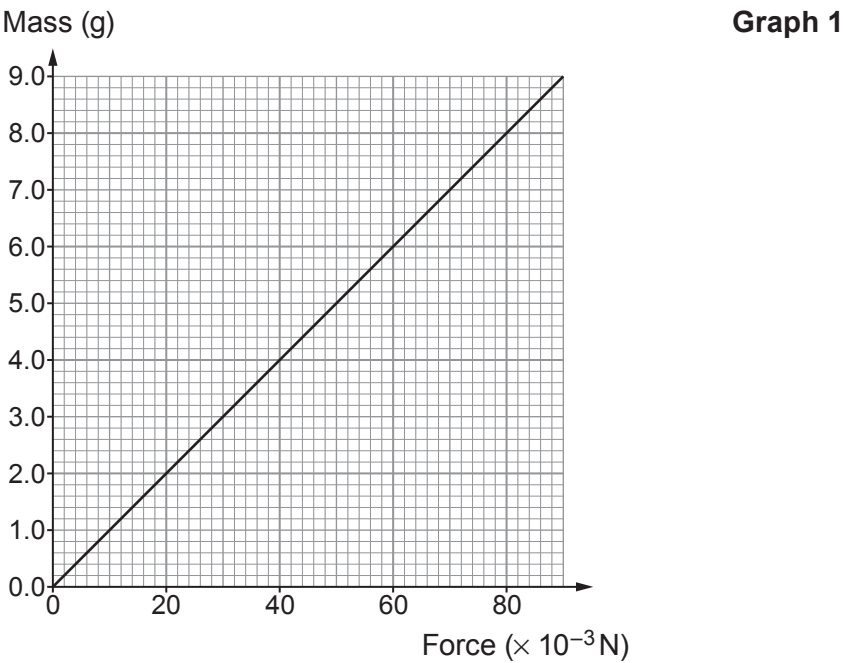
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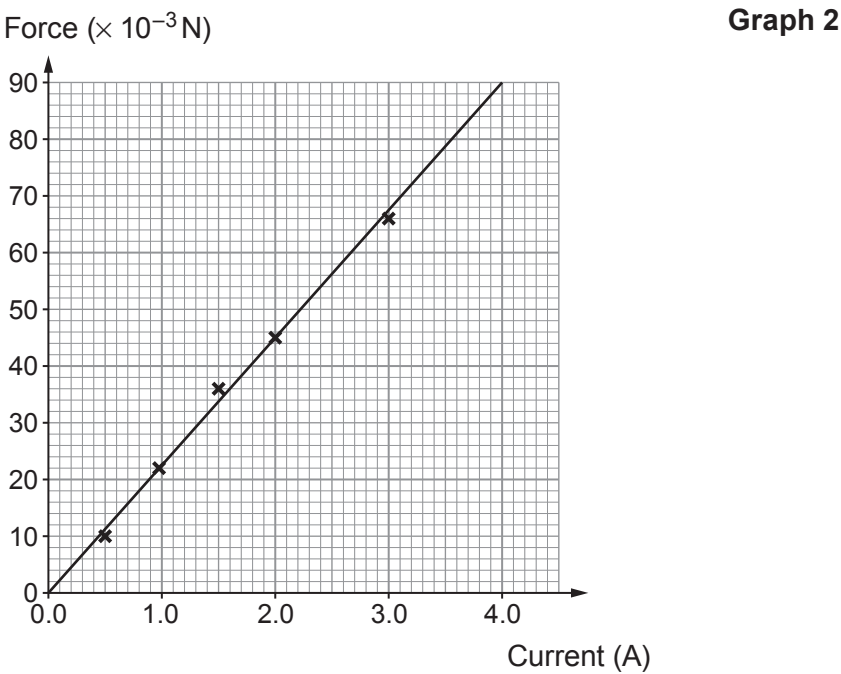
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- (b) When **no** current flows through the wire the digital balance is adjusted to read zero. The mass reading displayed on the digital balance is converted to a force, in newtons, using **Graph 1**.



The current through the wire is varied and the readings of force calculated. The data is plotted on the grid below and a best fit straight line is drawn.



Examiner  
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- (i) During the experiment a reading of **4.0 g** is displayed on the digital balance. Use the graphs opposite to complete the table. [2]

| Mass<br>(g) | Force<br>(N) | Current<br>(A) |
|-------------|--------------|----------------|
| 4.0         | .....        | .....          |

- (ii) Describe the relationship between current and force. [2]

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- (iii) A student correctly states that the line on **Graph 2** must obey the equation of a straight line:  $y = mx + c$ . Comparing this with  $F = BIl$  and, given that force is plotted on the  $y$ -axis, identify the **quantities** that represent the gradient and the intercept. [2]

Gradient = .....

Intercept = .....

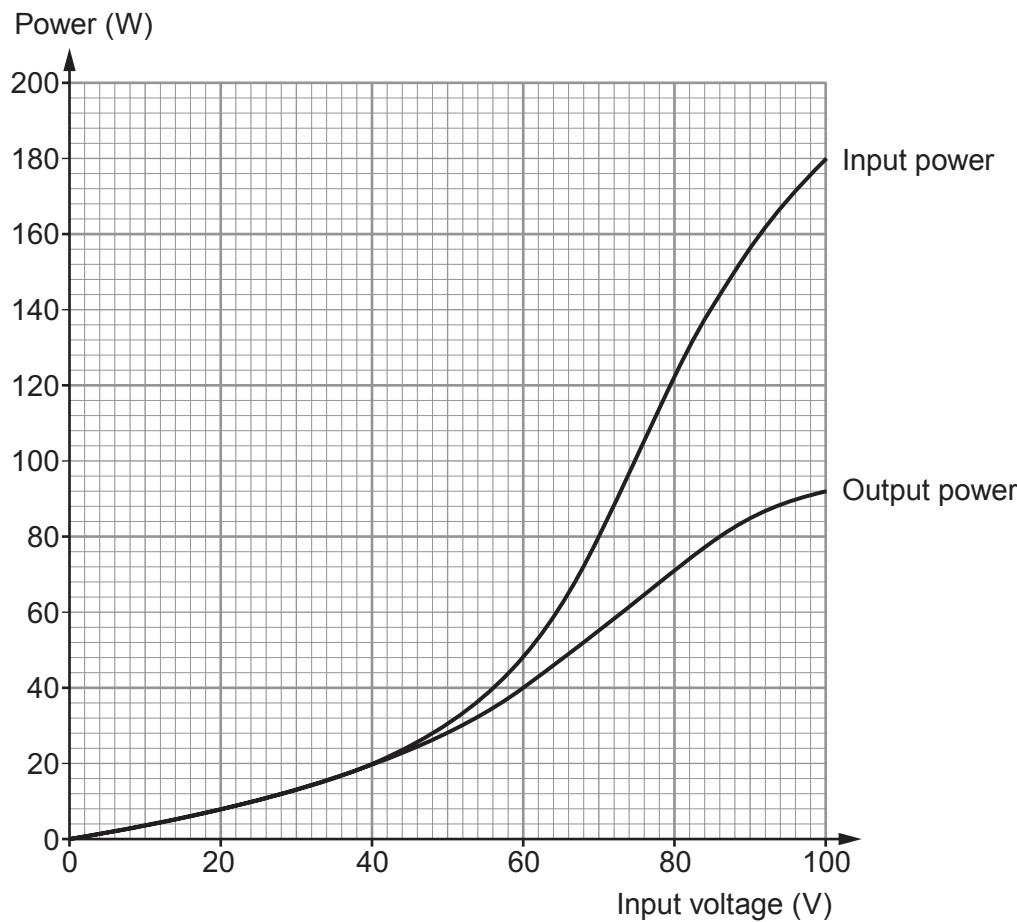
- (iv) The length of wire contained in the magnetic field,  $B$ , is 5.0 cm. Use information from **Graph 2** to calculate the magnetic field strength of the magnet used in the experiment. [3]

Magnetic field strength,  $B =$  ..... T

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8. Abigail carries out an investigation to compare the input power and output power of a transformer. Data is collected at different input voltages and displayed on a graph.



- (a) Abigail concludes that as the input voltage increases the transformer becomes less efficient. Use information on the graph to explain whether you agree or disagree with her. [2]

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- (b) Use an equation from page 2 to calculate the current supplied to the **input coil** of the transformer at 100V. [2]

Current = ..... A





Examiner  
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(c) Explain the purpose of the laminations in the soft iron core in a transformer. [2]

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(d) Explain the advantage of using high voltages for the transmission of electricity in the National Grid. [2]

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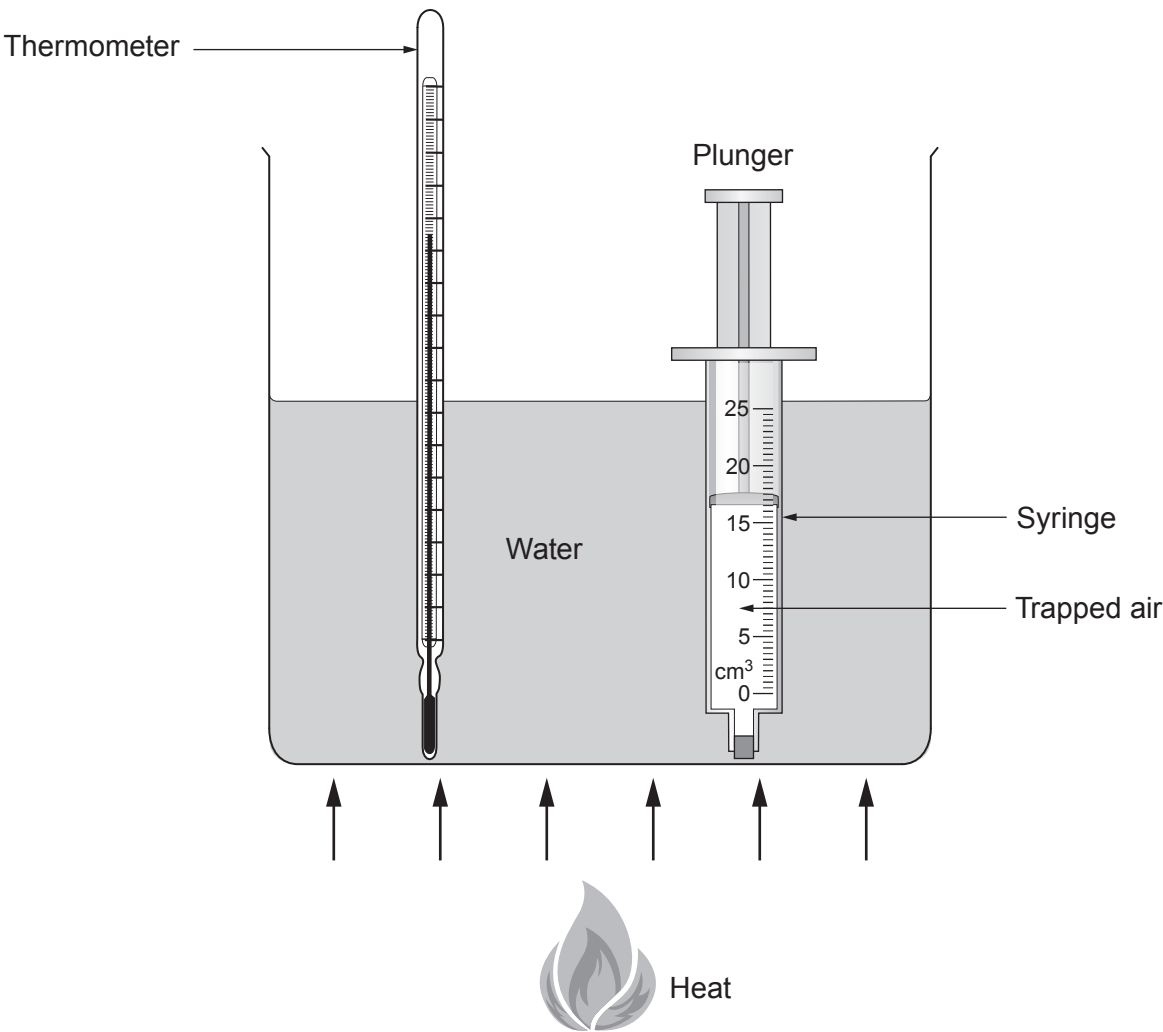
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9. In a class experiment the volume of a fixed mass of air changes when it is heated.



The air is trapped in a syringe which has been sealed at one end but the plunger at the other end is free to move in or out. The maximum volume the syringe can measure is 25.0 cm<sup>3</sup>. The volume of trapped air in the syringe is measured as the temperature of the water is increased. The results are shown in the table below.

| Temperature (°C) | Volume of trapped air (cm <sup>3</sup> ) |
|------------------|------------------------------------------|
| 10               | 19.5                                     |
| 30               | 21.0                                     |
| 50               | 22.5                                     |
| 60               | 23.0                                     |
| 80               | 24.5                                     |

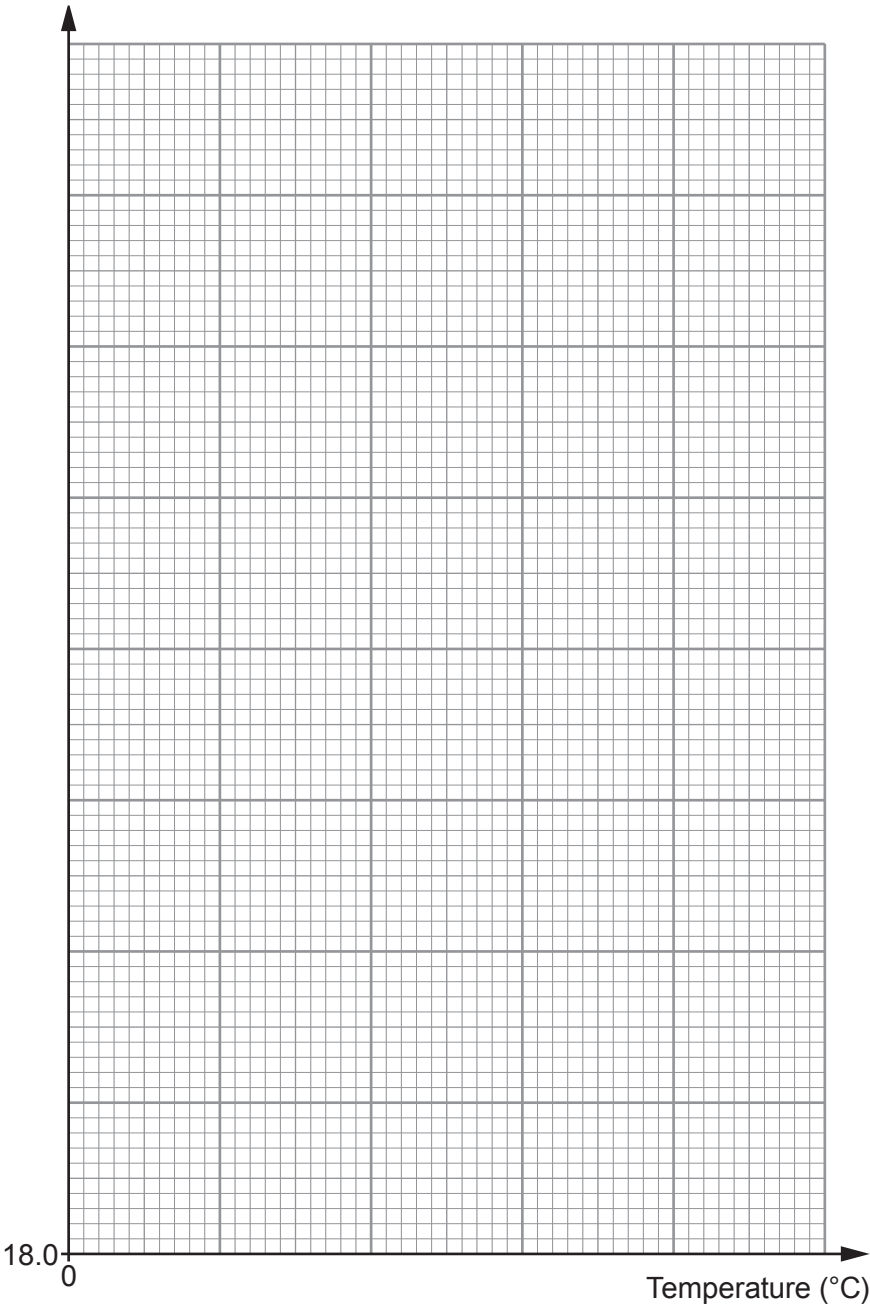


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- (a) Plot the data on the grid below and draw a suitable line.  
(Note that the scale on the volume axis does not start at zero.)

[3]

Volume of trapped air (cm<sup>3</sup>)



- (b) Explain, using your graph, whether the students can take a volume reading from the syringe at 100 °C.

[2]

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**QUESTION CONTINUES ON NEXT PAGE**



Examiner  
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(c) Explain how, if at all, the density of the trapped air changes as it is heated. [2]

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(d) Explain, in terms of the motion of molecules, why the volume of the trapped air increases as the temperature increases. [2]

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(e) The data in the table shows when the temperature increases from 10 °C to 60 °C the volume of trapped air increases from 19.5 cm<sup>3</sup> to 23.0 cm<sup>3</sup>. Using only this information, calculate the temperature, in °C, at which the volume of the trapped air is zero. [3]

Temperature = ..... °C

(f) State the name given to the temperature when the volume is zero. [1]

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END OF PAPER



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| Surname       | Centre Number | Candidate Number |
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GCSE  
3420UA0-1



MONDAY, 20 JUNE 2022 – MORNING

PHYSICS – Unit 1:  
Electricity, Energy and Waves

HIGHER TIER  
1 hour 45 minutes

| For Examiner's use only |              |              |
|-------------------------|--------------|--------------|
| Question                | Maximum Mark | Mark Awarded |
| 1.                      | 7            |              |
| 2.                      | 13           |              |
| 3.                      | 11           |              |
| 4.                      | 7            |              |
| 5.                      | 6            |              |
| 6.                      | 15           |              |
| 7.                      | 10           |              |
| 8.                      | 11           |              |
| Total                   | 80           |              |

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.  
You may use a pencil for graphs and diagrams only.  
Write your name, centre number and candidate number in the spaces at the top of this page.  
Answer **all** questions.  
Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.  
The assessment of the quality of extended response (QER) will take place in question 5.



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## Equations

|                                                                                                                                                                                      |                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| current = $\frac{\text{voltage}}{\text{resistance}}$                                                                                                                                 | $I = \frac{V}{R}$                              |
| total resistance in a series circuit                                                                                                                                                 | $R = R_1 + R_2$                                |
| total resistance in a parallel circuit                                                                                                                                               | $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$  |
| energy transferred = power $\times$ time                                                                                                                                             | $E = Pt$                                       |
| power = voltage $\times$ current                                                                                                                                                     | $P = VI$                                       |
| power = current <sup>2</sup> $\times$ resistance                                                                                                                                     | $P = I^2R$                                     |
| % efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$                                                            |                                                |
| density = $\frac{\text{mass}}{\text{volume}}$                                                                                                                                        | $\rho = \frac{m}{V}$                           |
| units used (kWh) = power (kW) $\times$ time (h)<br>cost = units used $\times$ cost per unit                                                                                          |                                                |
| wave speed = wavelength $\times$ frequency                                                                                                                                           | $v = \lambda f$                                |
| speed = $\frac{\text{distance}}{\text{time}}$                                                                                                                                        |                                                |
| pressure = $\frac{\text{force}}{\text{area}}$                                                                                                                                        | $p = \frac{F}{A}$                              |
| $p$ = pressure<br>$V$ = volume<br>$T$ = kelvin temperature                                                                                                                           | $\frac{pV}{T} = \text{constant}$               |
|                                                                                                                                                                                      | $T / \text{K} = \theta / ^\circ\text{C} + 273$ |
| change in thermal energy = mass $\times$ specific heat capacity $\times$ change in temperature                                                                                       | $\Delta Q = mc\Delta\theta$                    |
| thermal energy for a change of state = mass $\times$ specific latent heat                                                                                                            | $Q = mL$                                       |
| force on a conductor (at right angles to a magnetic field) carrying a current = magnetic field strength $\times$ current $\times$ length                                             | $F = BIl$                                      |
| $V_1$ = voltage across the primary coil<br>$V_2$ = voltage across the secondary coil<br>$N_1$ = number of turns on the primary coil<br>$N_2$ = number of turns on the secondary coil | $\frac{V_1}{V_2} = \frac{N_1}{N_2}$            |

## SI multipliers

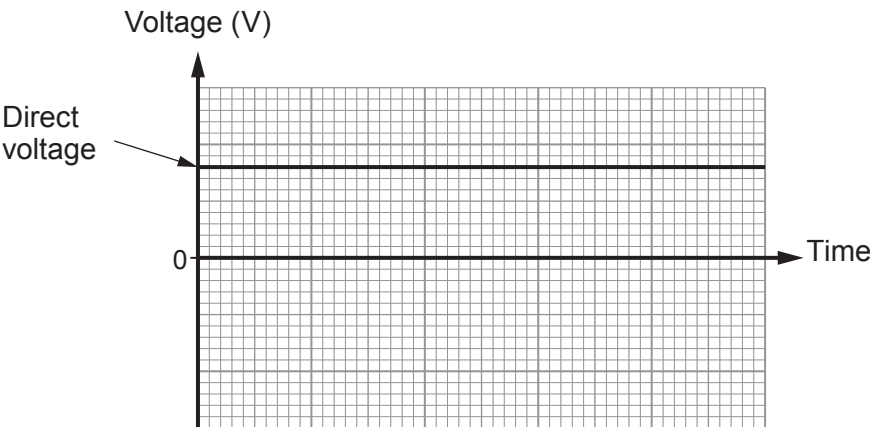
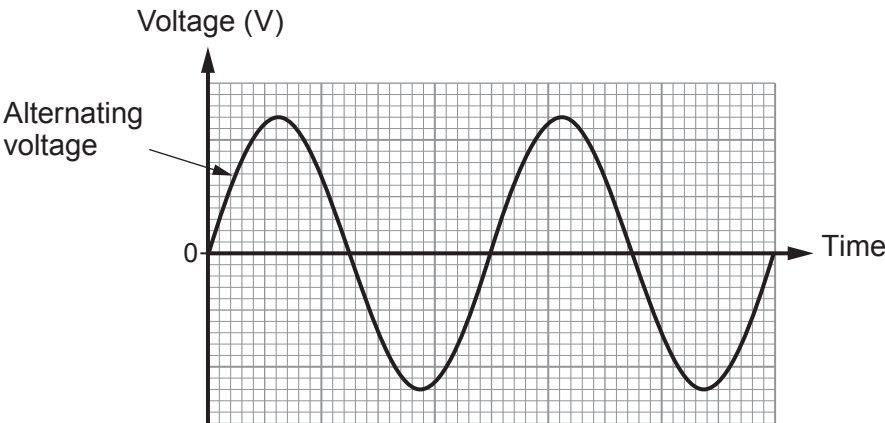
| Prefix | Multiplier          |
|--------|---------------------|
| p      | $1 \times 10^{-12}$ |
| n      | $1 \times 10^{-9}$  |
| $\mu$  | $1 \times 10^{-6}$  |
| m      | $1 \times 10^{-3}$  |

| Prefix | Multiplier         |
|--------|--------------------|
| k      | $1 \times 10^3$    |
| M      | $1 \times 10^6$    |
| G      | $1 \times 10^9$    |
| T      | $1 \times 10^{12}$ |



Answer **all** questions.

1. (a) The mains electricity supply to homes is at a voltage of 230 V a.c.  
In contrast to this, batteries supply d.c.  
These are each shown in the diagrams below.



An alternating voltage changes in size, but a direct voltage keeps the same size.  
State **one** other difference between the two.

[1]



- (b) Household circuits may be protected by a series of devices. These include:
- a miniature circuit breaker (mcb)
  - a fuse in a plug
  - a residual current circuit breaker (rccb).

The list below gives three situations in which one or another of these devices stop the power from being supplied.

1. An electric lawn mower cuts through the cable.
2. Too much current flows in a ring main because too many appliances are being used at the same time.
3. The live wire in a kettle touches its neutral wire.

- (i) Select and explain one situation above that puts the **user** in danger of an electric shock **and** which one of the devices should be used to keep the user safe. [2]

.....

.....

.....

- (ii) State **two** advantages of an mcb over a fuse in a household circuit. [2]

1. ....
2. ....

- (c) Describe the function of the live and neutral wires in household circuits. [2]

Live wire: .....

.....

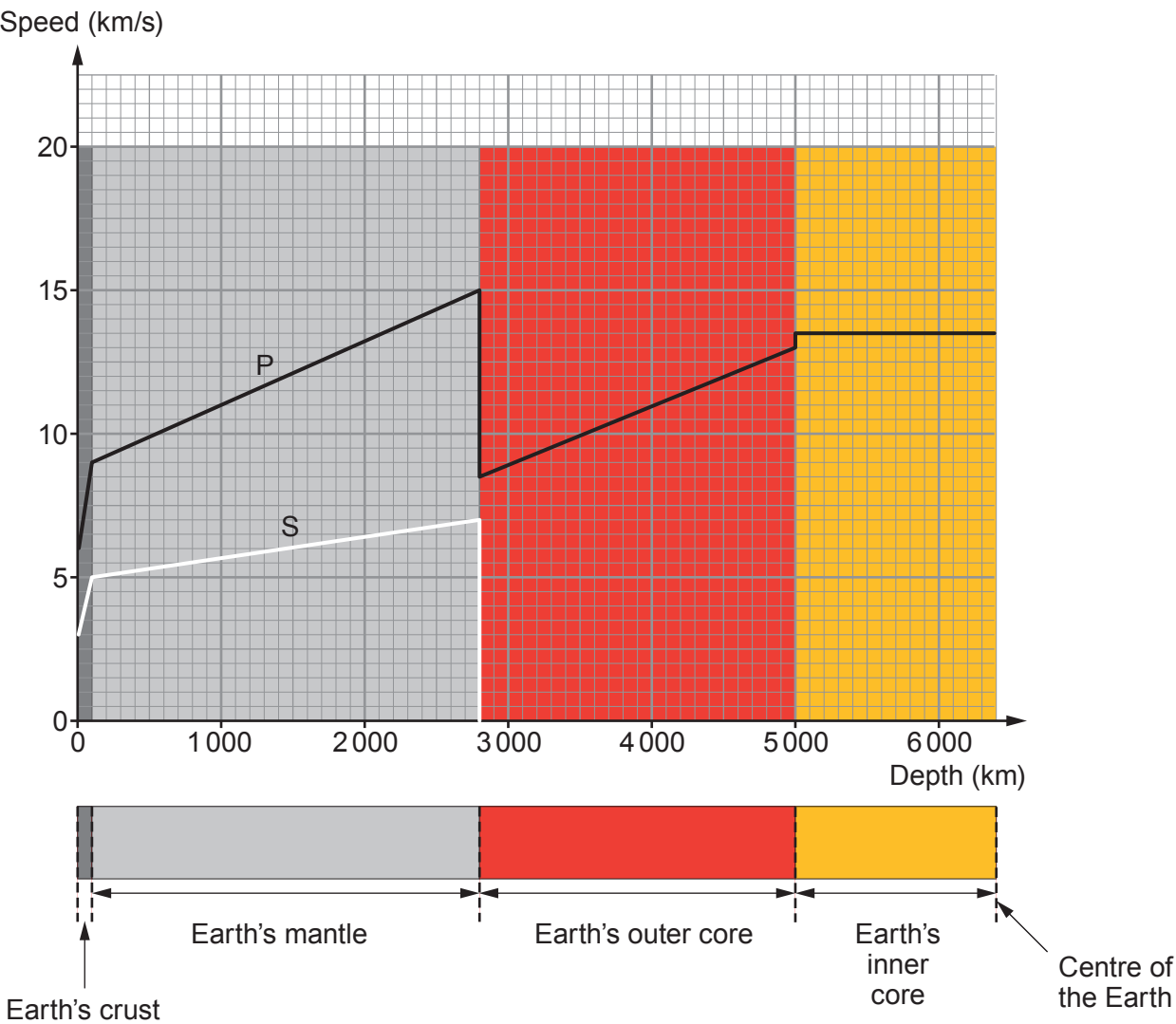
Neutral wire: .....

.....





2. P and S waves are produced during an earthquake. They travel at different speeds. The graph shows how the speed of P and S waves change with depth below the Earth's surface.



- (a) P waves are longitudinal waves. Explain what is meant by a longitudinal wave. [2]

.....

.....

- (b) Use the graph to calculate the difference in speed between the P wave and the S wave at a depth of **100 km**. [2]

Speed difference = ..... km/s



- (c) (i) Bob claims that the maximum speed of P waves is greater in solids than in liquids. Explain, using data from the graph, whether Bob's claim is correct. [2]

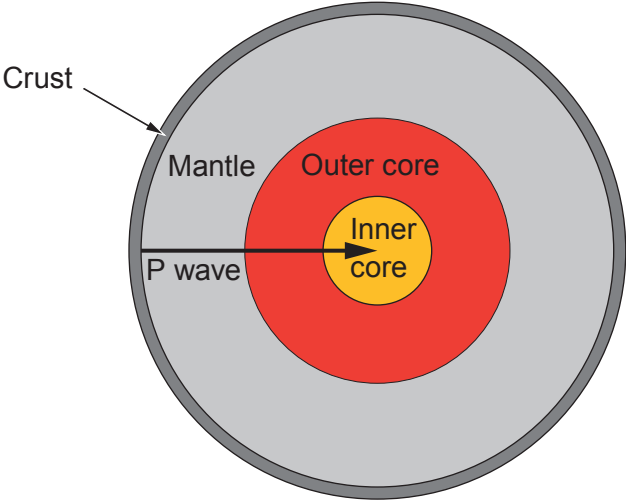
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- (ii) A P wave (shown in Diagram 1) travels a distance of 6 300 km in a straight line to the centre of the Earth in a time of 550 s.

Diagram 1



Bob states that the mean speed of the P wave over this distance is equal to its actual speed only at the depth of 3 500 km. Use an equation from page 2 to show whether Bob's statement is correct. [3]  
Space for calculations.

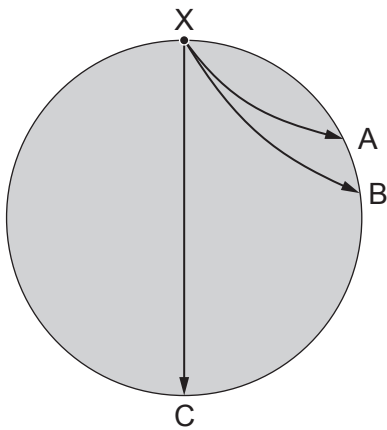
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(d) Diagram 2 shows waves passing through the Earth following an earthquake at point X.

Diagram 2



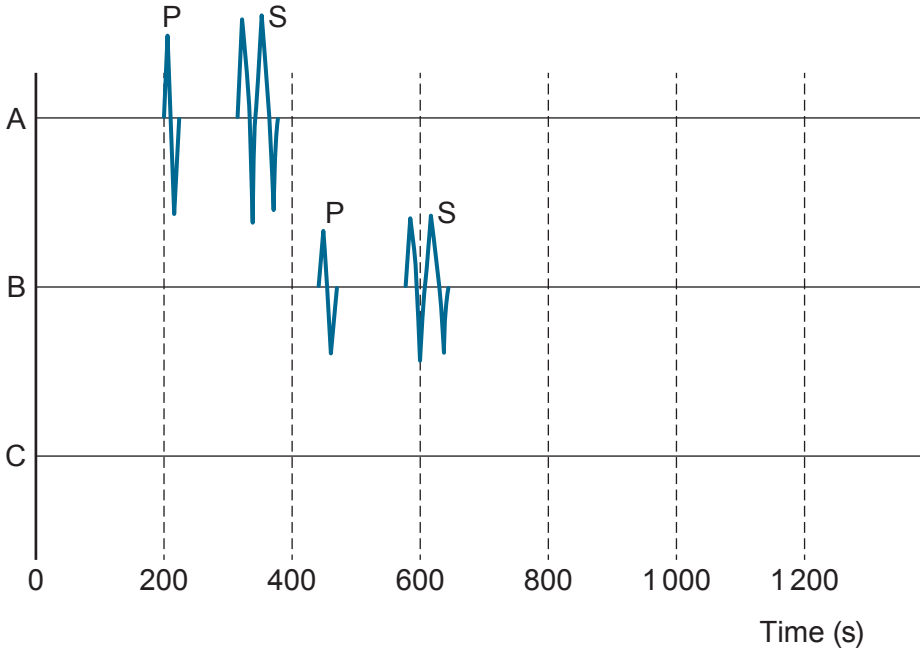
Monitoring stations at A, B and C detect the earthquake by signals that they receive.

- (i) Use the time given in part (c)(ii) to state the time taken for the P wave to travel from **X to C** (ignore the Earth's crust in this). [1]

Time = ..... s

- (ii) The signals received at monitoring stations A and B are shown in Diagram 3. **Add to Diagram 3** the signal that you would expect to be received at station C. [3]

Diagram 3



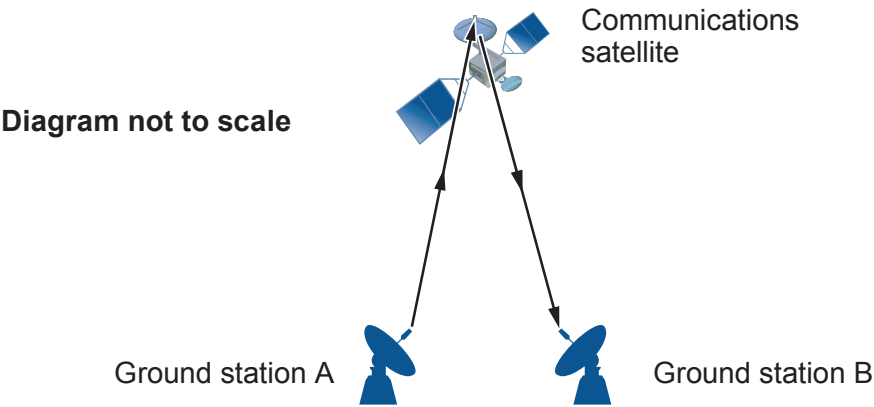
3. This question is about geostationary and geosynchronous satellites.

- (a) **Place a tick (✓) or a cross (×) in each box** in the table below.  
One box has been completed for you.

[4]

|                                                     | Geosynchronous<br>satellite | Geostationary<br>satellite |
|-----------------------------------------------------|-----------------------------|----------------------------|
| Stays above the same point<br>on Earth at all times | ×                           |                            |
| Orbits Earth once in 24 hours                       |                             |                            |
| Orbits above the equator                            |                             |                            |
| Must pass over the North<br>pole and South pole     |                             |                            |

- (b) The diagram shows a communications satellite relaying a signal between ground stations A and B.



- (i) Satellite communication uses microwaves of wavelength of 0.08 m.  
Use the equation:

$$\text{wave speed} = \text{frequency} \times \text{wavelength}$$

to calculate the frequency of the microwaves and state its unit.  
(Speed of light,  $c = 3 \times 10^8 \text{ m/s}$ )

[4]

Frequency = .....

Unit = .....



Examiner  
only

- (ii) The time delay between transmitting the signal at A and receiving it back on Earth at B is 0.24 s.  
Use an equation from page 2 to calculate the distance from the satellite to the ground stations. [3]  
(Speed of light,  $c = 3 \times 10^8 \text{ m/s}$ )

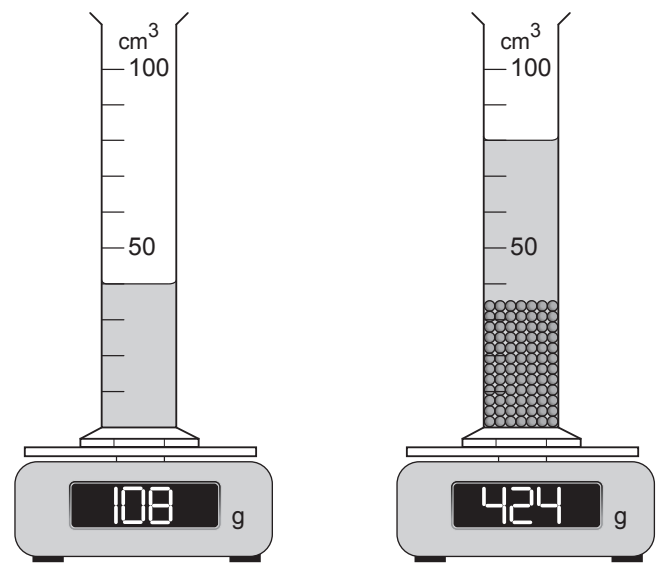
Distance = ..... m

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4. A group of students investigate density using an electronic balance, a measuring cylinder, water and some small steel ball bearings.
- They set the balance to zero and place a measuring cylinder on to the balance.**
- They then add water at 10 cm<sup>3</sup> intervals and record both the mass and volume.
- At some point they start adding the steel ball bearings instead of water, and record the mass and volume each time the water level rises by 10 cm<sup>3</sup>.
- They continue until the final volume of 100 cm<sup>3</sup> is reached.
- The arrangement is shown below.



The students produce a table of their results, which is shown below.

| Volume (cm <sup>3</sup> ) | Mass (g) |
|---------------------------|----------|
| 0                         | 68       |
| 10                        | 78       |
| 20                        | 88       |
| 30                        | 98       |
| 40                        | 108      |
| 50                        | 187      |
| 60                        | 266      |
| 70                        | 345      |
| 80                        | 424      |
| 90                        | 503      |
| 100                       | 582      |



Examiner  
only

- (a) One member of the group claims that water has a density value of  $1\text{ g/cm}^3$ . Explain whether her claim is correct. [No calculations are required.] [1]

.....

.....

- (b) (i) State the volume at which they start adding the steel ball bearings. [1]

Volume = .....  $\text{cm}^3$

- (ii) Give a reason for your answer. [1]

.....

.....

- (c) Use an equation from page 2 to calculate the density of steel. [2]

Density of steel = .....  $\text{g/cm}^3$

- (d) State **two** improvements that could be made to this experiment that would lead to a more accurate value for the density of steel. [2]

1. ....
2. ....

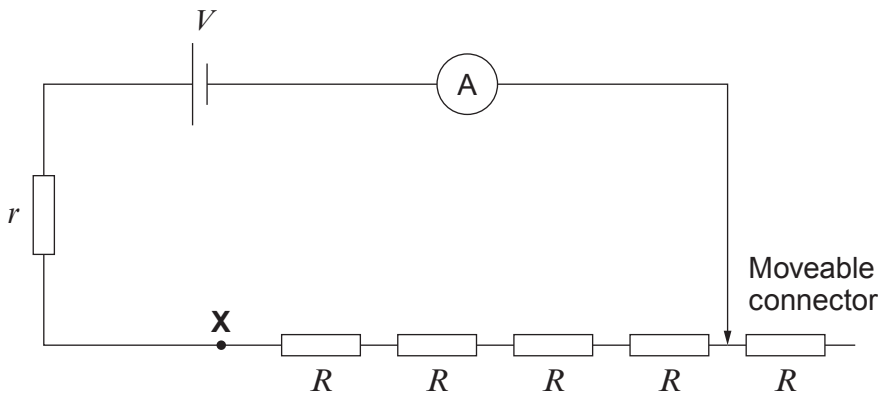
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6. A student sets up the following experiment to find how current is related to the number of identical resistors,  $R$ , in the circuit. She uses a power supply that provides a constant voltage,  $V$ , and includes a safety resistor,  $r$ , in the circuit as shown.



- (a) The moveable connector is moved to point  $X$  in the circuit. Explain why the ammeter might be damaged if there was no safety resistor,  $r$ , included in the circuit. [2]

.....

.....

.....

- (b) The student takes a series of readings of current as the number of resistors in the circuit is increased. Her results are shown below.

| Number of resistors, $R$ | Total circuit resistance ( $\Omega$ ) | Current, $I$ (A) |                |       | $\frac{1}{\text{current}}$ ( $\frac{1}{\text{A}}$ ) |
|--------------------------|---------------------------------------|------------------|----------------|-------|-----------------------------------------------------|
|                          |                                       | First reading    | Second reading | Mean  |                                                     |
| 0                        | 10                                    | 1.200            | 1.200          | 1.200 | 0.83                                                |
| 1                        | 30                                    | 0.390            | 0.410          | ..... | 2.5                                                 |
| 2                        | 50                                    | 0.230            | 0.250          | 0.240 | 4.2                                                 |
| 4                        | 90                                    | 0.133            | 0.133          | 0.133 | 7.5                                                 |
| 5                        | 110                                   | 0.110            | 0.108          | 0.109 | 9.2                                                 |



Examiner  
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Refer to the table to answer the following questions.

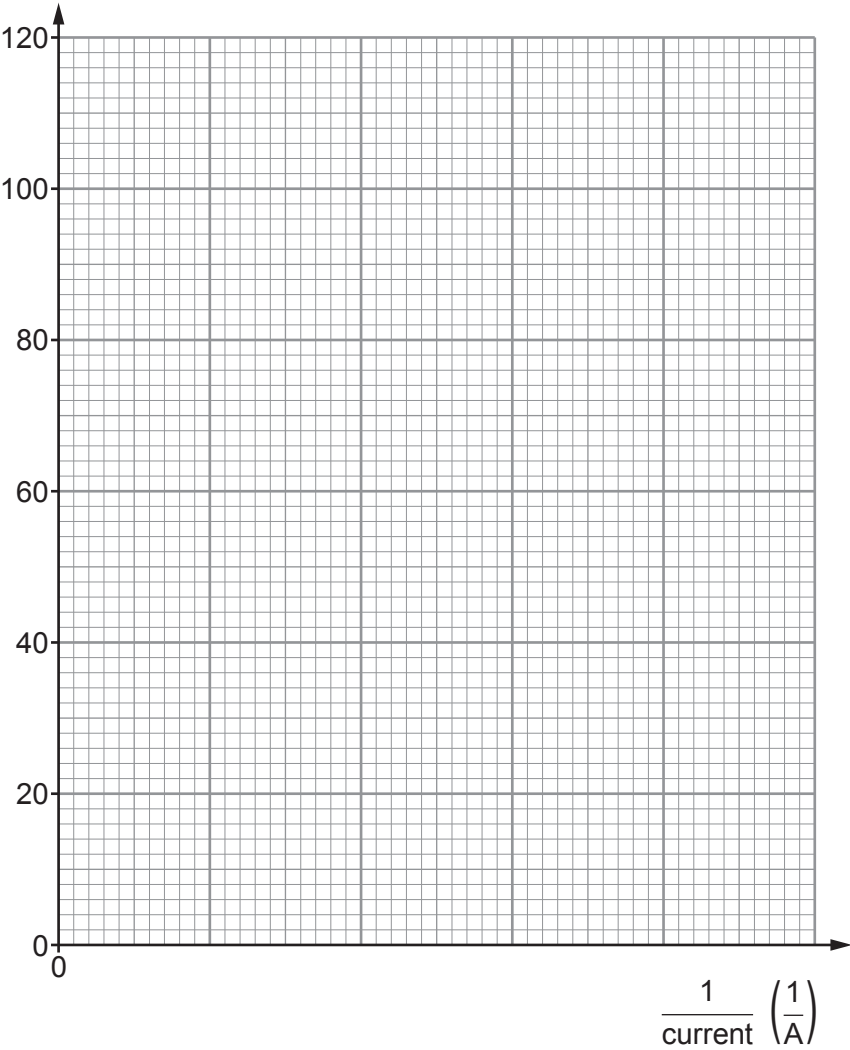
(i) **Complete** the missing mean value in the table. [1]  
Space for calculation.

(ii) State the value of the resistance of the safety resistor,  $r$ . [1]  
Resistance = .....  $\Omega$

(iii) Determine the value of the resistance of each resistor,  $R$ . [1]  
Resistance = .....  $\Omega$

(iv) Plot the results on the grid below and draw a suitable line. [3]

Total circuit resistance ( $\Omega$ )



Examiner  
only

- (c) Use your answer to part (b)(iii) and the graph to determine the current when 3 of the resistors labelled  $R$  are included in the circuit. [3]

Current = ..... A

- (d) (i) The gradient of the graph gives the value of the voltage of the battery. Calculate its value and show your workings. [2]

Voltage = ..... V

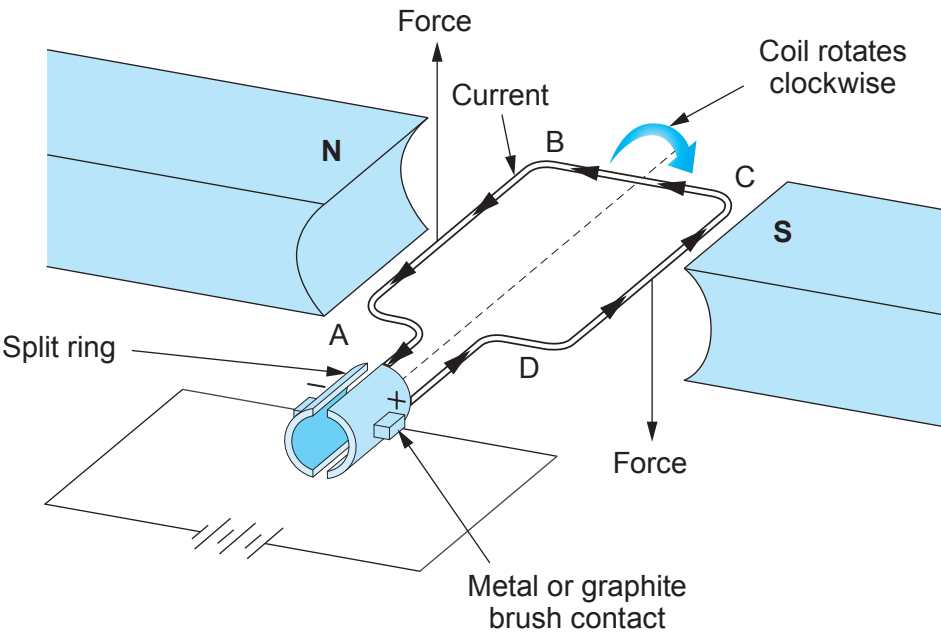
- (ii) Describe **one** other way that the battery voltage could be calculated. [2]  
No calculations are required.

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7. The diagram shows a simple d.c. electric motor consisting of a single turn of wire ABCD.



(a) Describe how Fleming's left-hand rule can be used to show that the side **AB** of the wire moves upwards. [3]

.....

.....

.....

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(b) By considering the current in **AB**, explain how the split ring allows the coil to continue to rotate. [2]

.....

.....

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Examiner  
only

(c) The sides AB and CD have lengths of 6.0 cm and the side BC has a length of 3.0 cm. The magnetic field has a strength of 0.04 T and a current of 0.7 A passes through the coil.

(i) Use the equation:

force on a conductor (at right angles to a magnetic field) carrying a current = magnetic field strength × current × length

to calculate the force on side **AB** of the coil. [3]

Force = ..... N

(ii) Describe the effect on the force, on side **AB** of the coil, of doubling **both** the battery voltage **and** the number of coils. [2]

.....

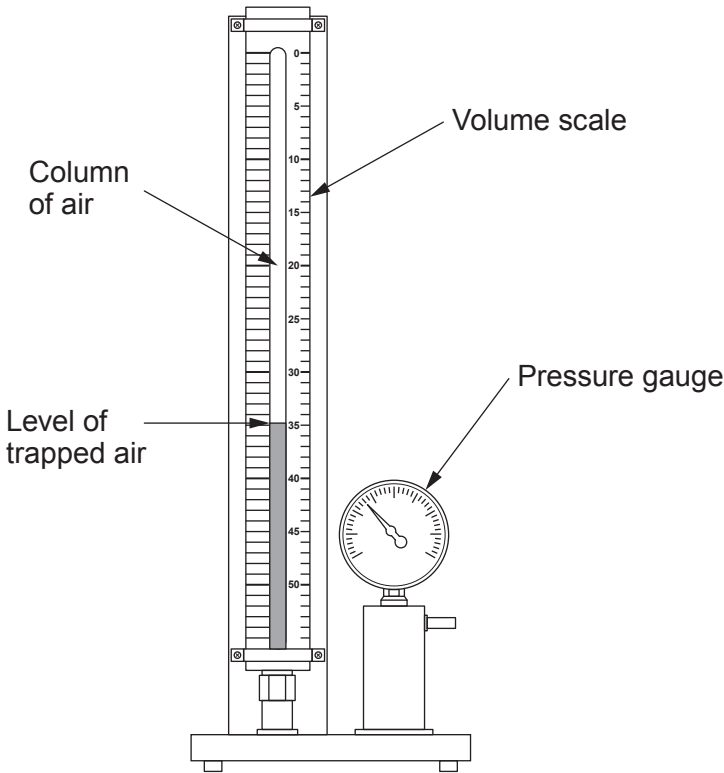
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8. (a) The volume of a gas is related to its pressure.  
A mass of air is contained in a glass tube that is attached to a pressure gauge.



- (i) When the pressure shown by the pressure gauge increases and the temperature is constant, the volume of the column of air decreases.  
Describe the effect, if any, on the motion of the molecules and on the collisions with the container. [2]

.....

.....

.....

- (ii) At a pressure of  $1.2 \times 10^5 \text{ Pa}$ , the volume is  $18 \text{ cm}^3$ .  
Explain, without calculation, what happens to the volume when the pressure is increased to  $2.4 \times 10^5 \text{ Pa}$  at constant temperature. [2]

.....

.....

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Examiner  
only

- (b) In another experiment, air initially at 27 °C is kept at constant volume. The air pressure is increased from  $2.4 \times 10^5 \text{ Pa}$ , to  $3.0 \times 10^5 \text{ Pa}$ .

(i) State the initial temperature in Kelvin. [1]

Temperature = ..... K

(ii) Use an equation from page 2 to calculate the **change in temperature** of the air. [3]

Change in temperature = ..... K

- (c) A mass of  $5 \times 10^{-3} \text{ kg}$  of air requires 105.5 J of thermal energy to increase its temperature by 21 °C.

Use the equation:

$$\Delta Q = mc\Delta\theta$$

to calculate the specific heat capacity,  $c$ , of air. [3]

Specific heat capacity = ..... J/kg °C

END OF PAPER

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|               |               |                  |
|---------------|---------------|------------------|
| Surname       | Centre Number | Candidate Number |
| First name(s) |               | 0                |



GCSE  
3420UA0-1



MONDAY, 19 JUNE 2023 – AFTERNOON

PHYSICS – Unit 1:  
Electricity, Energy and Waves

HIGHER TIER  
1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.  
The assessment of the quality of extended response (QER) will take place in question 3.

| For Examiner’s use only |              |              |
|-------------------------|--------------|--------------|
| Question                | Maximum Mark | Mark Awarded |
| 1.                      | 7            |              |
| 2.                      | 13           |              |
| 3.                      | 6            |              |
| 4.                      | 11           |              |
| 5.                      | 7            |              |
| 6.                      | 11           |              |
| 7.                      | 14           |              |
| 8.                      | 11           |              |
| Total                   | 80           |              |



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## Equations

|                                                                                                                                                                                      |                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| current = $\frac{\text{voltage}}{\text{resistance}}$                                                                                                                                 | $I = \frac{V}{R}$                              |
| total resistance in a series circuit                                                                                                                                                 | $R = R_1 + R_2$                                |
| total resistance in a parallel circuit                                                                                                                                               | $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$  |
| energy transferred = power $\times$ time                                                                                                                                             | $E = Pt$                                       |
| power = voltage $\times$ current                                                                                                                                                     | $P = VI$                                       |
| power = current <sup>2</sup> $\times$ resistance                                                                                                                                     | $P = I^2 R$                                    |
| % efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$                                                            |                                                |
| density = $\frac{\text{mass}}{\text{volume}}$                                                                                                                                        | $\rho = \frac{m}{V}$                           |
| units used (kWh) = power (kW) $\times$ time (h)<br>cost = units used $\times$ cost per unit                                                                                          |                                                |
| wave speed = wavelength $\times$ frequency                                                                                                                                           | $v = \lambda f$                                |
| speed = $\frac{\text{distance}}{\text{time}}$                                                                                                                                        |                                                |
| pressure = $\frac{\text{force}}{\text{area}}$                                                                                                                                        | $p = \frac{F}{A}$                              |
| $p$ = pressure $V$ = volume $T$ = kelvin temperature                                                                                                                                 | $\frac{pV}{T} = \text{constant}$               |
|                                                                                                                                                                                      | $T / \text{K} = \theta / ^\circ\text{C} + 273$ |
| change in thermal energy = mass $\times$ specific heat capacity $\times$ change in temperature                                                                                       | $\Delta Q = mc\Delta\theta$                    |
| thermal energy for a change of state = mass $\times$ specific latent heat                                                                                                            | $Q = mL$                                       |
| force on a conductor (at right angles to a magnetic field) carrying a current = magnetic field strength $\times$ current $\times$ length                                             | $F = BIl$                                      |
| $V_1$ = voltage across the primary coil<br>$V_2$ = voltage across the secondary coil<br>$N_1$ = number of turns on the primary coil<br>$N_2$ = number of turns on the secondary coil | $\frac{V_1}{V_2} = \frac{N_1}{N_2}$            |

## SI multipliers

| Prefix | Symbol | Conversion factor             | Multiplier          |
|--------|--------|-------------------------------|---------------------|
| pico   | p      | divide by 1 000 000 000 000   | $1 \times 10^{-12}$ |
| nano   | n      | divide by 1 000 000 000       | $1 \times 10^{-9}$  |
| micro  | $\mu$  | divide by 1 000 000           | $1 \times 10^{-6}$  |
| milli  | m      | divide by 1000                | $1 \times 10^{-3}$  |
| centi  | c      | divide by 100                 | $1 \times 10^{-2}$  |
| kilo   | k      | multiply by 1000              | $1 \times 10^3$     |
| mega   | M      | multiply by 1 000 000         | $1 \times 10^6$     |
| giga   | G      | multiply by 1 000 000 000     | $1 \times 10^9$     |
| terra  | T      | multiply by 1 000 000 000 000 | $1 \times 10^{12}$  |



Answer **all** questions.

1. The table below gives information about four types of power station.

The table ranks the power stations in order from 1 to 4 for three different features.  
Rank 1 is best and rank 4 is worst.

| Power station | Efficiency | Rank | Running cost     | Rank | Emissions                                                                   | Rank |
|---------------|------------|------|------------------|------|-----------------------------------------------------------------------------|------|
| Type A        | 25 %       | 4    | Second highest   | 3    | Highest polluting emissions                                                 | 4    |
| Type B        |            | 1    | Practically zero | 1    | No emissions                                                                | 1    |
| Type C        | 35 %       | 3    | Highest          | 4    | Has cleaner emissions than type A power stations                            | 2    |
| Type D        | 40 %       | 2    | Second lowest    | 2    | Cleaner emissions than type C power stations but produces radioactive waste | 3    |

(a) Use the information in the table to answer the following questions.

(i) Gareth says that the best type of power station to recommend overall by ranking is **type B**. Explain whether you agree with him. [2]

.....

.....

.....



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only

- (ii) The energy sources for different types of power station are **fossil fuel**, **nuclear** and **hydroelectric**.

**Complete the table** below for the energy sources for types **A**, **B**, **C** and **D**. [3]  
Each energy source may be used once, more than once, or not at all.

| Type     | Energy source |
|----------|---------------|
| <b>A</b> | .....         |
| <b>B</b> | .....         |
| <b>C</b> | .....         |
| <b>D</b> | .....         |

- (b) Use the information below and an equation from page 2 to calculate the % efficiency of a **type B** power station. [2]

Input energy = 200 000 MJ  
Heat energy produced = 30 000 MJ  
Electrical energy produced = 170 000 MJ

% efficiency = .....

|   |
|---|
|   |
| 7 |



2. Water waves refract when they travel from deep water to shallow water.

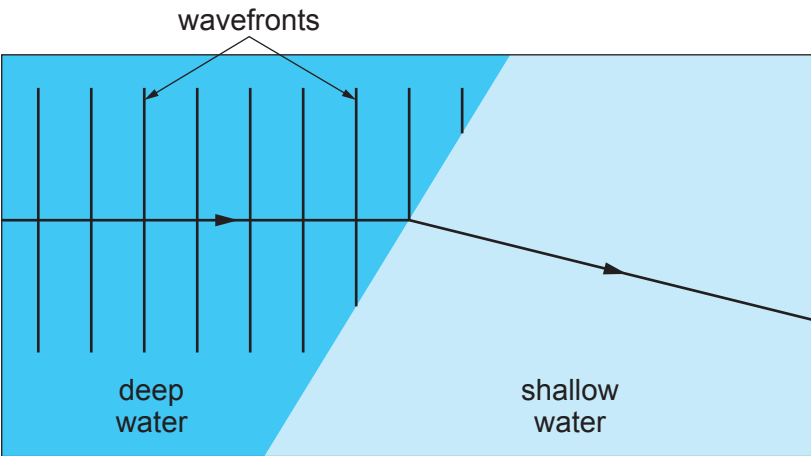
Refraction happens because the speed of the wave changes.

The wavelength also changes because the frequency is constant.

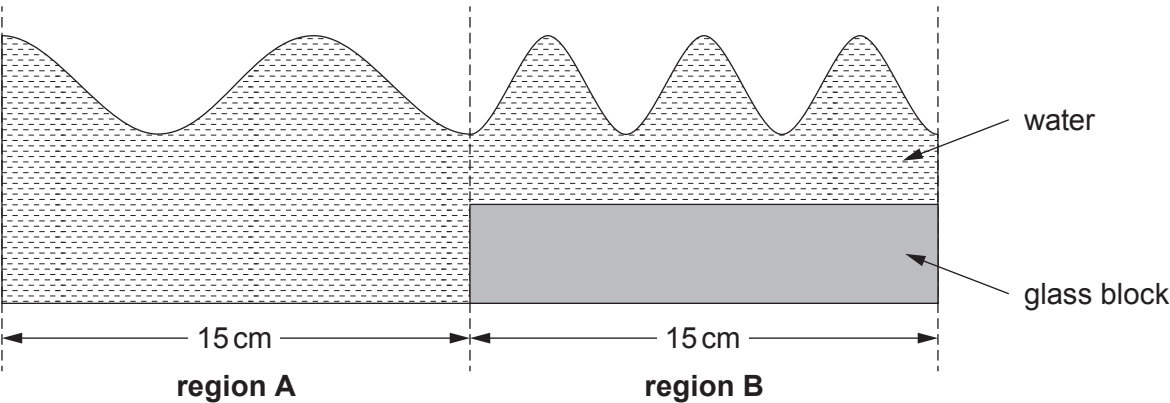
Students use a ripple tank to investigate this effect.

(a) Complete the diagram below to show the water waves in shallow water.

[3]



(b) The depth of the shallow water can be changed by using glass blocks of different thicknesses.



Regions A and B are both 15 cm long.

- (i) I. How many waves are shown in **region A**? ..... [1]
- II. Calculate the wavelength of the waves in **region A**. [1]

wavelength = ..... cm



(ii) John says that the wave speed in **region B** is greater than the wave speed in **region A**. Explain whether John is correct. [2]

.....

.....

.....

(c) In another experiment using a different tank, students investigate how the depth of water affects wave speed.

They change the depth of the water using different thickness glass blocks.

The water level is kept constant at 10 cm.

The table below shows their results.

| Thickness of glass block (cm) | Depth of water (cm) | Wave speed (cm/s) |
|-------------------------------|---------------------|-------------------|
| 8                             | 2                   | 60                |
| 6                             | 4                   | 75                |
| 4                             | .....               | 82                |

(i) **Complete the table.** [1]

(ii) Use the equation:

wavelength =  $\frac{\text{wave speed}}{\text{frequency}}$

to calculate the wavelength of water waves of frequency 50 Hz when the thickness of the glass block is 6 cm. [2]

wavelength = ..... cm

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(iii) Janet states that when the thickness of the glass block decreases by 2 cm the wave speed increases by a quarter.  
Explain to what extent Janet is correct.  
Space for calculations.

[3]

Examiner  
only

|    |
|----|
|    |
| 13 |



Examiner  
only

3. A saucepan is used to heat water on a gas cooker.



Explain, **in terms of particles**, the processes of conduction through the metal base of the saucepan **and** convection in the water. [6 QER]

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4. One type of petrol car costs £12 500 to buy.  
The equivalent model as an electric car costs £24 500.

- (a) The electric car travels 240 km on a full charge.  
It takes 8 hours to fully charge the electric car battery.  
A home charging point is rated at 7 kW.  
Homes are charged 30 p for each kWh of electricity used.

Use the information above and equations from page 2 to calculate the cost, **in £**, to travel 240 km.

[3]

charging cost for 240 km = £ .....

- (b) Fuel consumption for the petrol car is 15 km/l (kilometres per litre).  
The cost of petrol is £1.60 per litre.

- (i) Calculate the fuel cost if the petrol car is driven 240 km.

[2]

fuel cost for 240 km = £ .....





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(ii) Both cars are driven 14 400 km per year.

I. Calculate the difference in running costs for one year. [2]

difference in running costs per year = £ .....

II. Calculate the payback time of the **extra cost** if the electric car is bought instead of the petrol car. [2]

payback time = ..... years

(c) It is often claimed that electric cars are environmentally friendly because they do not produce greenhouse gases when used. Explain whether you agree. [2]

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5. The table below gives examples of transverse and longitudinal waves.

| Transverse waves | Longitudinal waves |
|------------------|--------------------|
| light            | sound              |
| S waves          | P waves            |

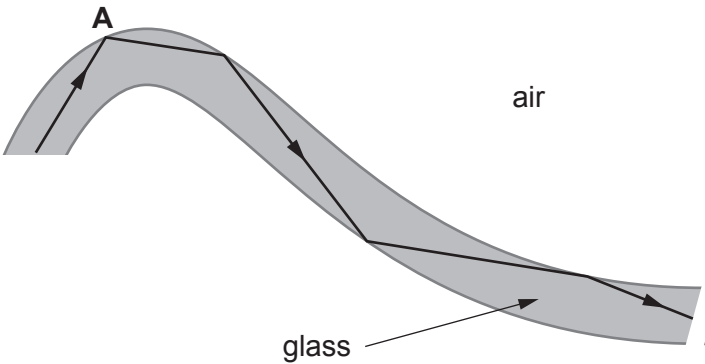
(a) Describe the difference between transverse and longitudinal waves. [2]

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(b) Light waves can travel through an optical fibre in the way shown in the diagram.



(i) Name the effect where light changes direction as shown at point **A**. [1]

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(ii) Explain why light waves travel through the fibre as shown. [2]

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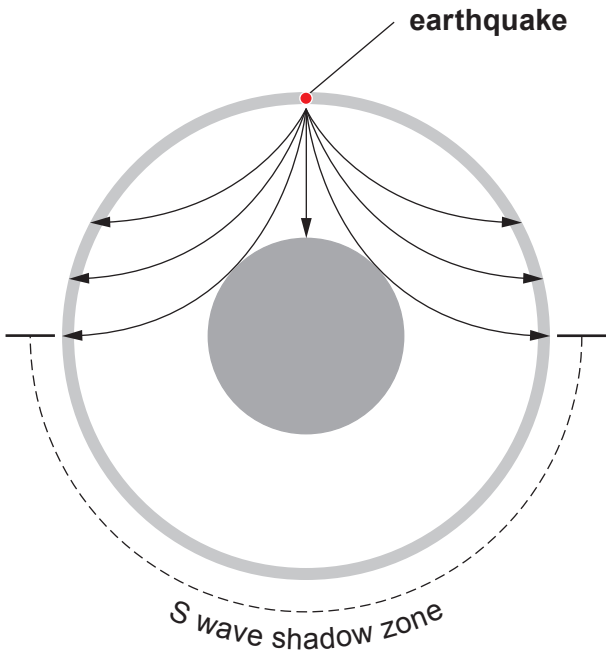
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(c) The diagram below shows an S wave shadow zone.



Explain how this led to the development of a model for the structure of the Earth. [2]

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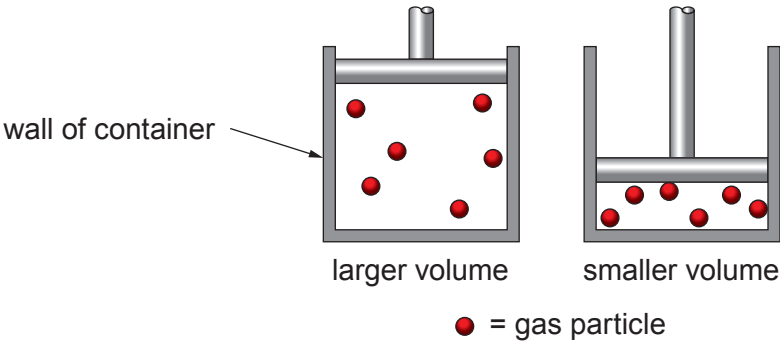
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6. Kinetic theory is used to explain the behaviour of gases.

- (a) A fixed mass of gas is kept at a constant temperature in a container. Explain why the pressure increases when its volume is decreased.

[3]



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- (b) Students are given data about the behaviour of a fixed mass of gas kept at **constant pressure**. The volume of the gas changes with temperature as shown in the table below.

| Temperature (°C) | Volume (cm <sup>3</sup> ) |
|------------------|---------------------------|
| –223             | 20                        |
| –173             | 40                        |
| –73              | 80                        |



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(i) Chris states that

$$\frac{\text{volume}}{\text{temperature}} = \text{constant}$$

when temperatures are measured in °C. Explain whether the results agree. [2]  
Space for calculations.

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(ii) The students plot a graph of the results opposite. Explain how they can use the graph to determine the value of absolute zero in °C. [2]

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(c) A balloon is filled to a volume of 2800 cm<sup>3</sup> at 7 °C (280 K).  
The balloon is heated to a temperature of 67 °C.  
The pressure remains constant.

Use the equation:

$$\frac{pV}{T} = \text{constant}$$

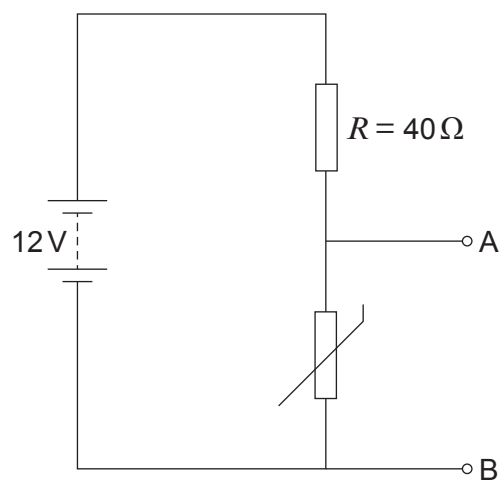
to calculate the new volume of the balloon. [4]  
[T (K) = θ (°C) + 273]

volume = ..... cm<sup>3</sup>

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7. The circuit below is used to investigate how the resistance of a thermistor changes as temperature increases.

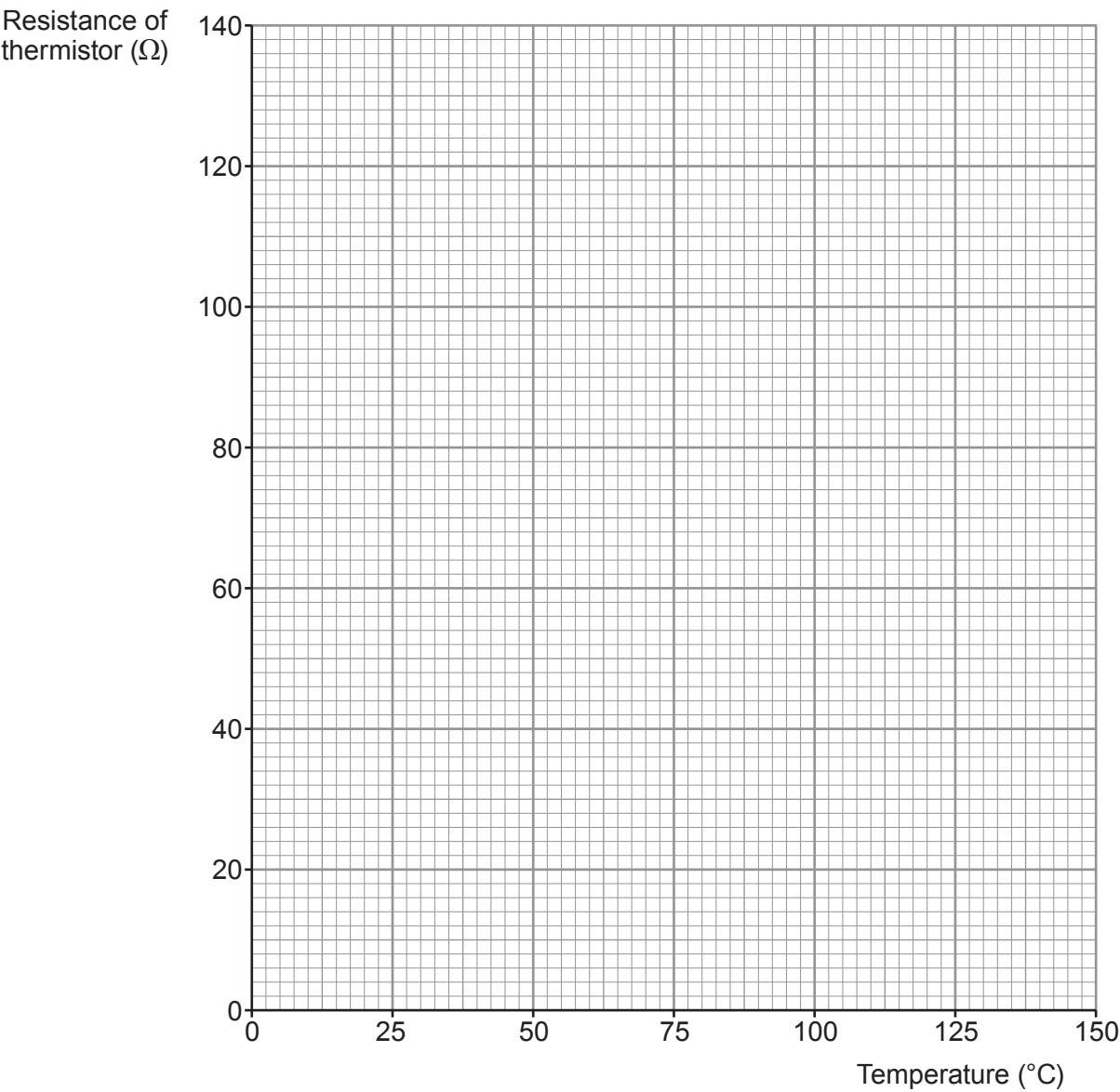


- (a) **Add an ammeter and a voltmeter** to the diagram so the necessary measurements can be taken. [2]
- (b) **Plot the results shown** below on the grid opposite and draw a curve of best fit. [2]

| Temperature (°C) | Resistance of thermistor (Ω) |
|------------------|------------------------------|
| 0                | 120                          |
| 25               | 92                           |
| 50               | 70                           |
| 75               | 52                           |
| 100              | 40                           |
| 125              | 30                           |



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- (c) The resistance of the resistor,  $R$ , remains constant at  $40\,\Omega$ .
- (i) Calculate the voltage across the terminals AB when the temperature is  $100\,^{\circ}\text{C}$ . [3]

voltage = ..... V



- (ii) Tom says as temperature increases the current increases.  
Explain whether you agree.

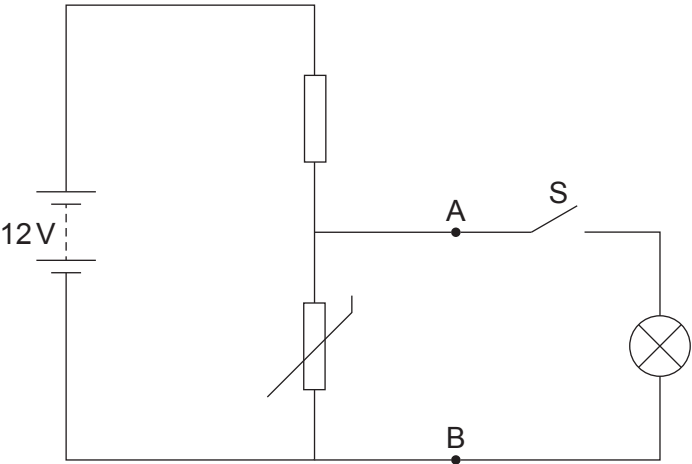
[2]

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- (d) A lamp and a switch are now connected across the terminals AB, as shown in the diagram below.



- (i) The lamp has a power of 3W at 12V.  
Use equations from page 2 to calculate the resistance of the lamp at this power and voltage.

[3]

resistance = .....  $\Omega$





(ii) The switch, S, is now closed. Explain, without calculation, what happens to the total resistance of the circuit. [2]

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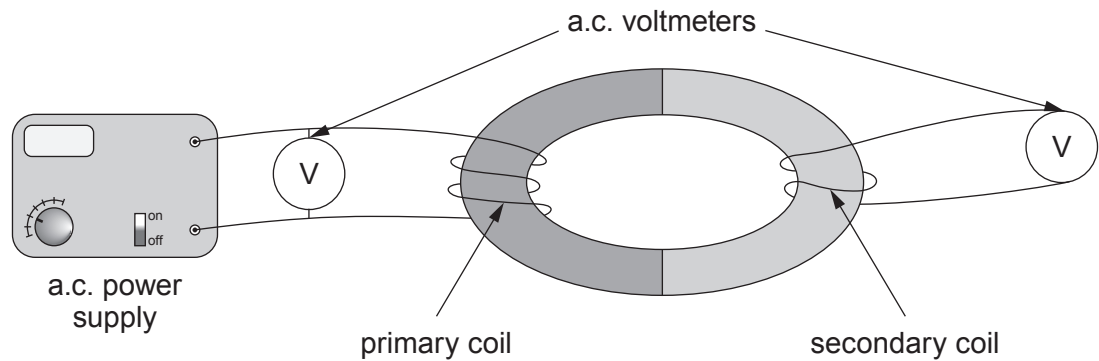
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8. Transformers are used to change the size of an a.c. voltage.

- (a) Describe how the apparatus in the diagram below is used to investigate the effect of the number of turns on the secondary coil on the output of a transformer. [4]



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- (b) Explain why transformers only change a.c. voltages. [2]

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- (c) Explain how step-up transformers increase the efficiency of the National Grid system. [2]

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- (d) A transformer is used to reduce the 230V mains voltage to 11.5V to run a television. The primary coil of the transformer has 600 turns. Use an equation from page 2 to calculate the number of turns on the secondary coil. [3]

number of turns = .....

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END OF PAPER

